



POLITECNICO
MILANO 1863

**ADVANCED DYNAMICS OF
MECHANICAL SYSTEMS**

Lab Assignment 1

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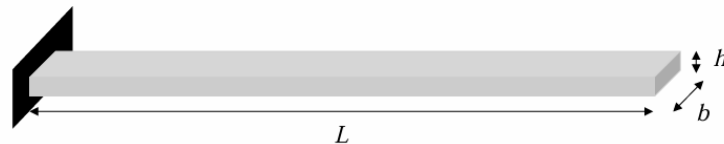
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Part A: Cantilever beam

Natural frequencies and mode shapes

The problem is about a clamped aluminum beam with a rectangular cross section and a free end.



Parameter	symbol	unit	value
Lenght	L	mm	1200
Thickness	h	mm	8
Width	b	mm	40
Density	ρ	kg/m ³	2700
Young's Modulus	E	GPa	68

The standing wave solution of the beam is the following one:

$$w(x, t) = [A \cos(\gamma x) + B \sin(\gamma x) + C \cosh(\gamma x) + D \sinh(\gamma x)] \cos(\omega t + \varphi)$$

In order to find the natural frequencies, it is necessary to compute the four **boundary conditions**, since there are four unknown constants.

Clamped end	Free end
No displacement $w(0, t) = 0$ No rotation $\frac{\partial w(x, t)}{\partial x} \Big _{x=0} = 0$	Moment equilibrium $EJ \frac{\partial^2 w(x, t)}{\partial x^2} \Big _{x=L} = 0$ Vertical force equilibrium $EJ \frac{\partial^3 w(x, t)}{\partial x^3} \Big _{x=L} = 0$

Then, using the boundary conditions, it is possible to determine the **matrix $H(\omega)$** .

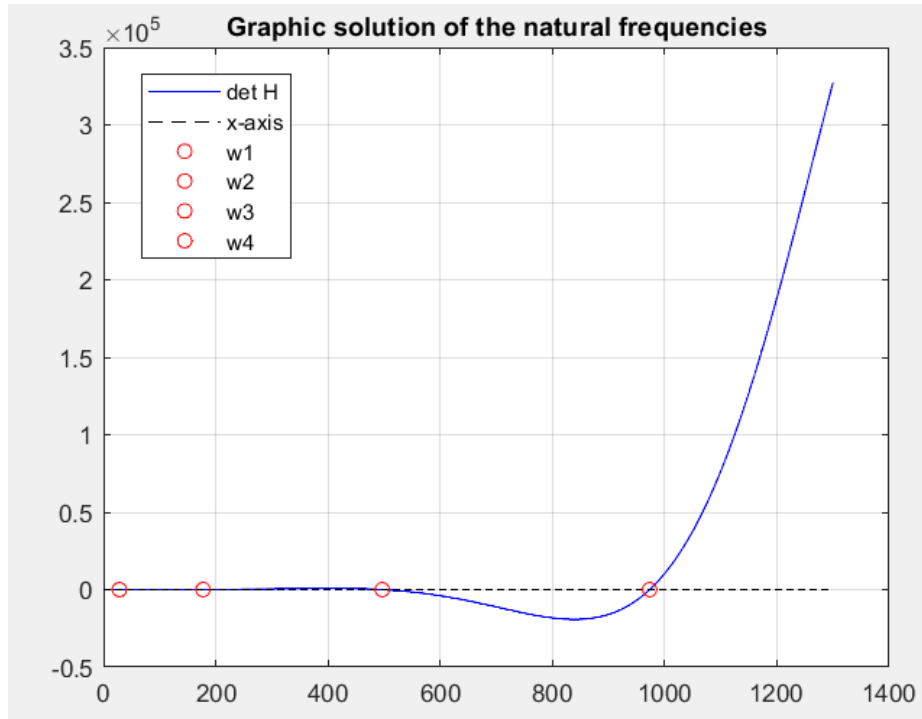
$$H(\omega) = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -\cos(\gamma L) & -\sin(\gamma L) & \cosh(\gamma L) & \sinh(\gamma L) \\ \sin(\gamma L) & -\cos(\gamma L) & \sinh(\gamma L) & \cosh(\gamma L) \end{bmatrix}$$

Considering that:

$$\gamma = \sqrt[4]{\frac{m\omega^2}{EJ}} \quad m = \rho b h \quad J = \frac{bh^3}{12}$$

- b and h are expressed in [m];
- E is expressed in [Pa];

After that, the first four **natural frequencies** ω_0 can be calculated plotting the function **det[H(ω)]** and using the graphic method. Given that the determinant of H in function of ω_0 is equal to 0, the zeros of the function are the natural frequencies.



Done that, **the constants** can be found substituting the natural frequencies in the matrix H and considering constant A equal to 1.

$$[H(\omega_i)] \underline{z}^{(i)} = \underline{0}$$

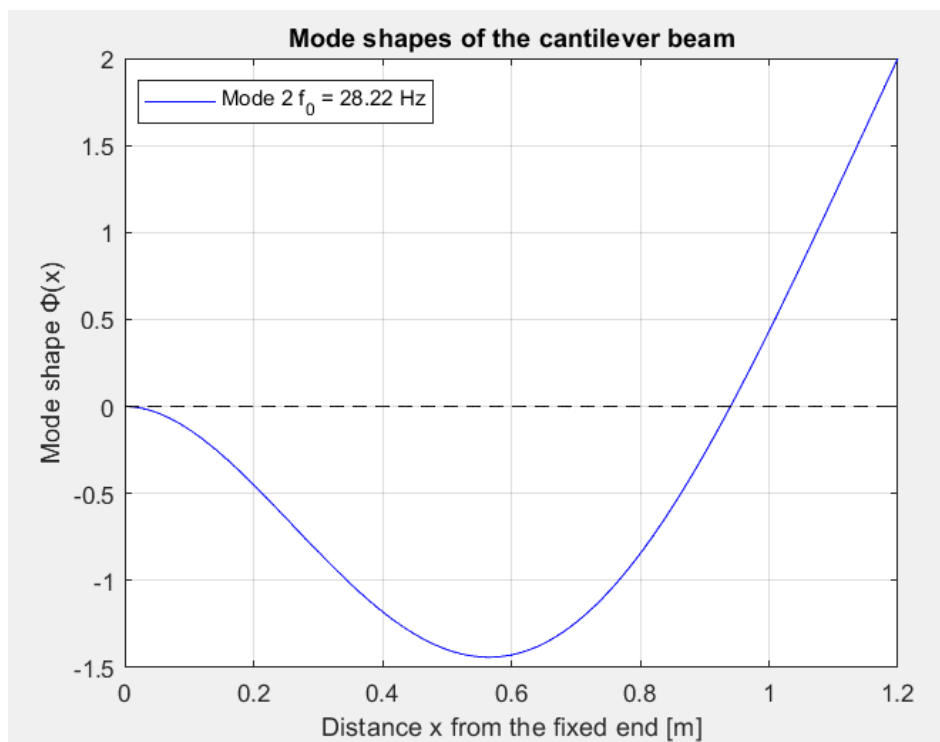
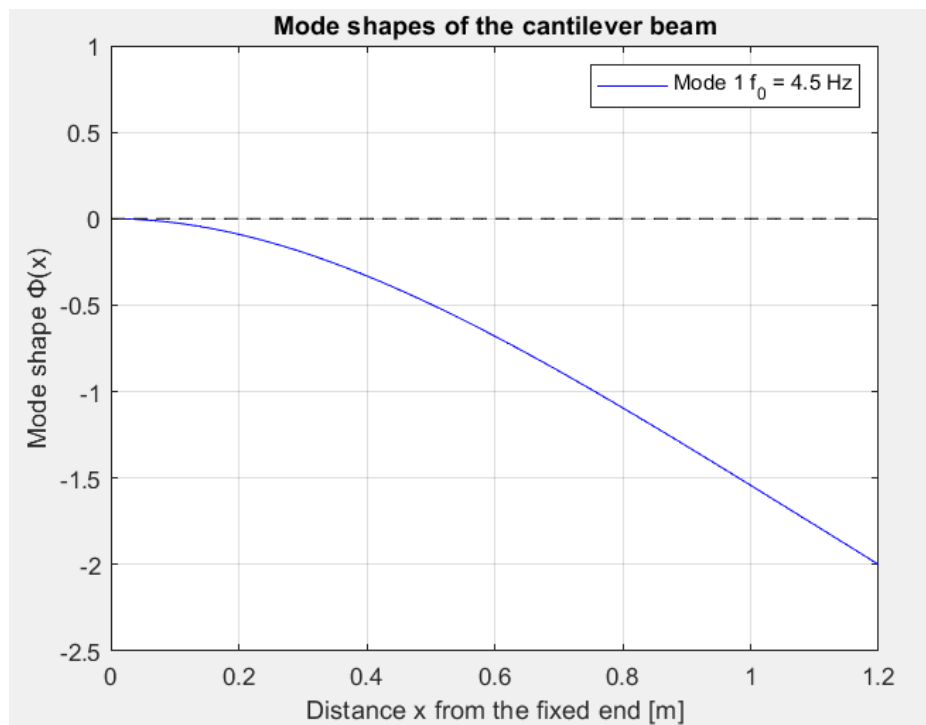
$$H(\omega_i) = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ -\cos(\gamma_i L) & -\sin(\gamma_i L) & \cosh(\gamma_i L) & \sinh(\gamma_i L) \\ \sin(\gamma_i L) & -\cos(\gamma_i L) & \sinh(\gamma_i L) & \cosh(\gamma_i L) \end{bmatrix}$$

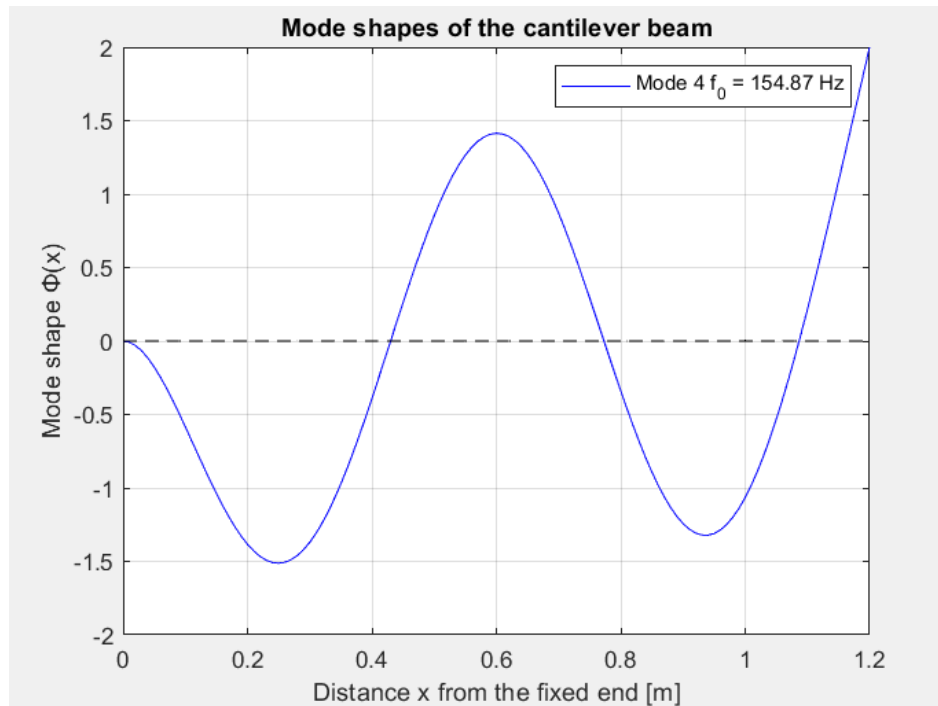
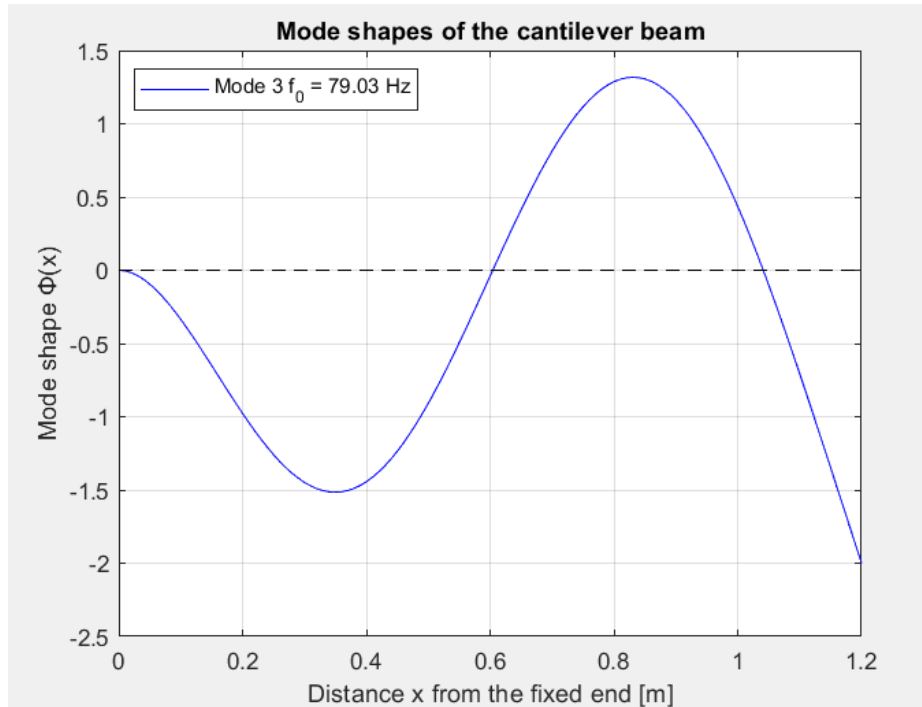
$\underline{N}^{(i)}$

$\hat{H}(\omega_i)$

$$\underline{\widehat{z}}^{(i)} = -[\hat{H}(\omega_i)]^{-1} \underline{N}^{(i)} \quad \underline{\widehat{z}}^{(i)} = \begin{Bmatrix} B_i \\ C_i \\ D_i \end{Bmatrix} \quad \underline{z}^{(i)} = \begin{Bmatrix} 1 \\ \underline{\widehat{z}}^{(i)} \end{Bmatrix} \quad i = 1, \dots, 4$$

Substituting the constants in the standing wave solution, it is possible to compute and plot the **mode shapes**.





Frequency Response Functions

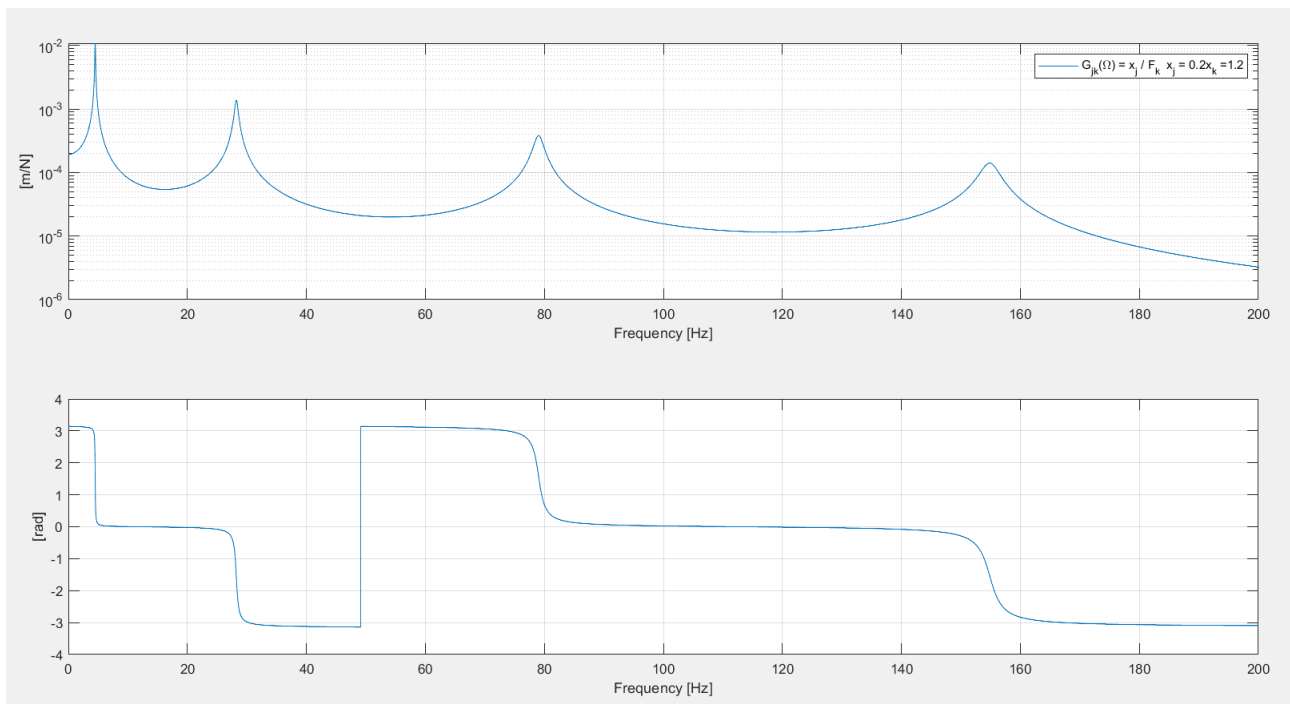
The FRFs have been computed through the following equations, taking account of four combinations of input and output positions and assuming the damping ratio ξ_i equal to 1%.

$$G_{jk}(\Omega) = \sum_{i=1}^n \frac{\Phi_i(x_j) \Phi_i(x_k)/m_i}{-\Omega^2 + j2\xi_i\omega_i\Omega + \omega_i^2} \quad m_i = \int_0^L m \Phi_i^2(x) dx$$

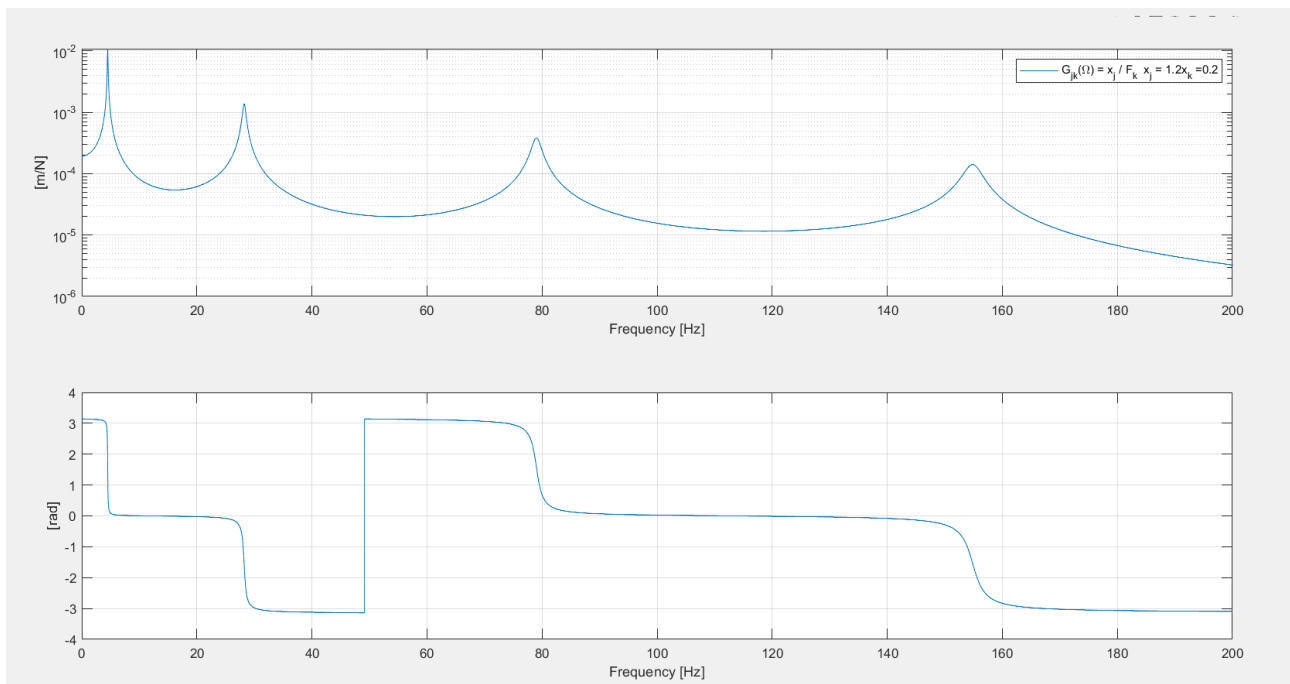
1st case.

Output $x_j = 0.2$ m

Input $x_k = 1.2$ m



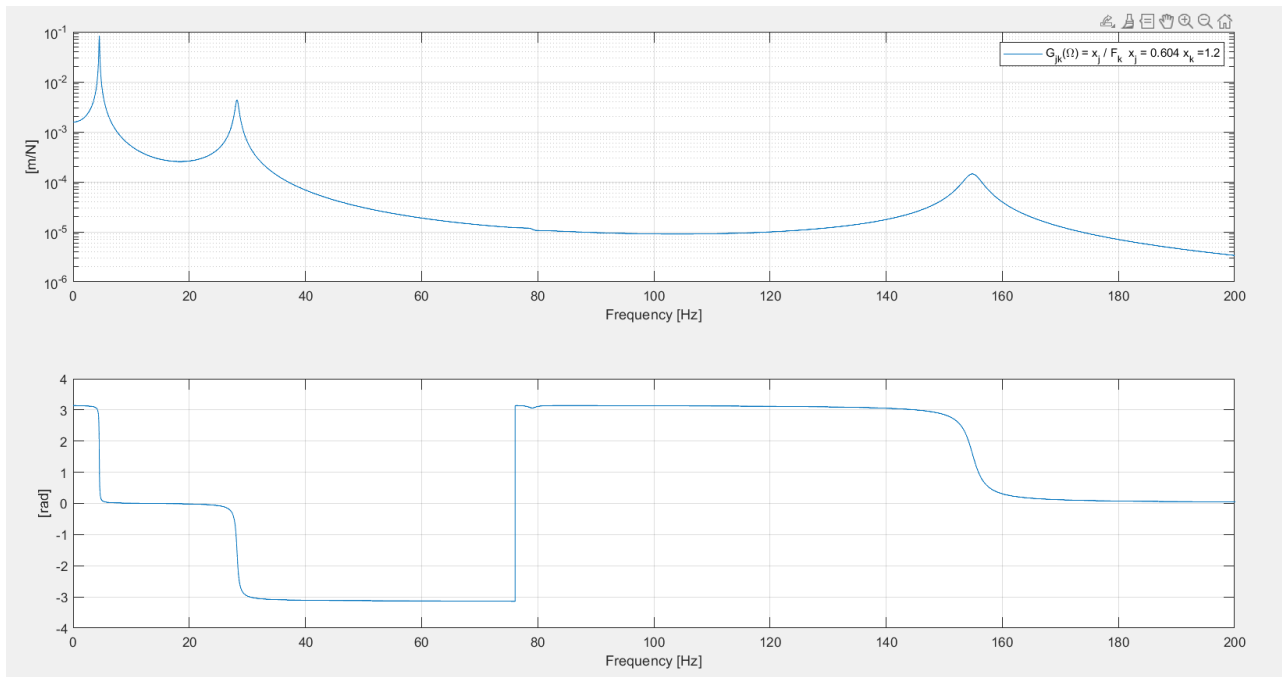
In order to verify the **reciprocity principle**, it was decided to switch the I/O positions.



2nd case: not observable mode

Output $x_j = 0.604$ m (node of the 3rd mode shape)

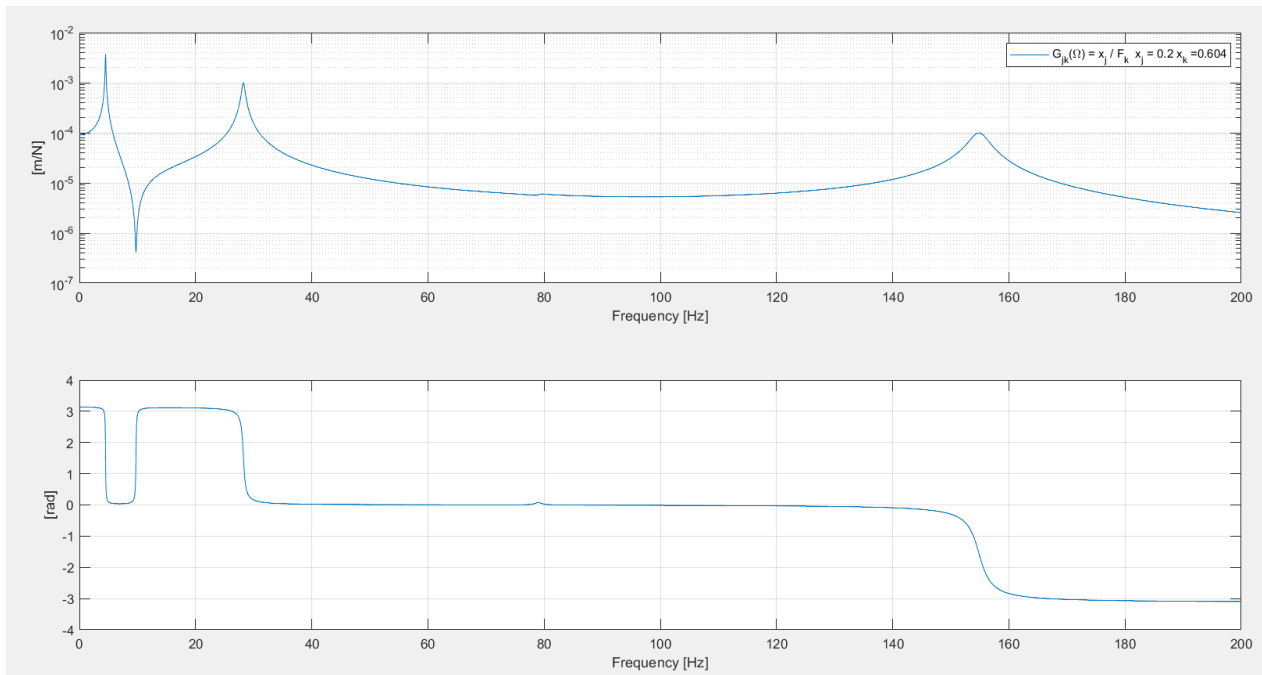
Input $x_k = 1.2$ m



3rd case: not controllable mode

Output $x_j = 0.2$ m

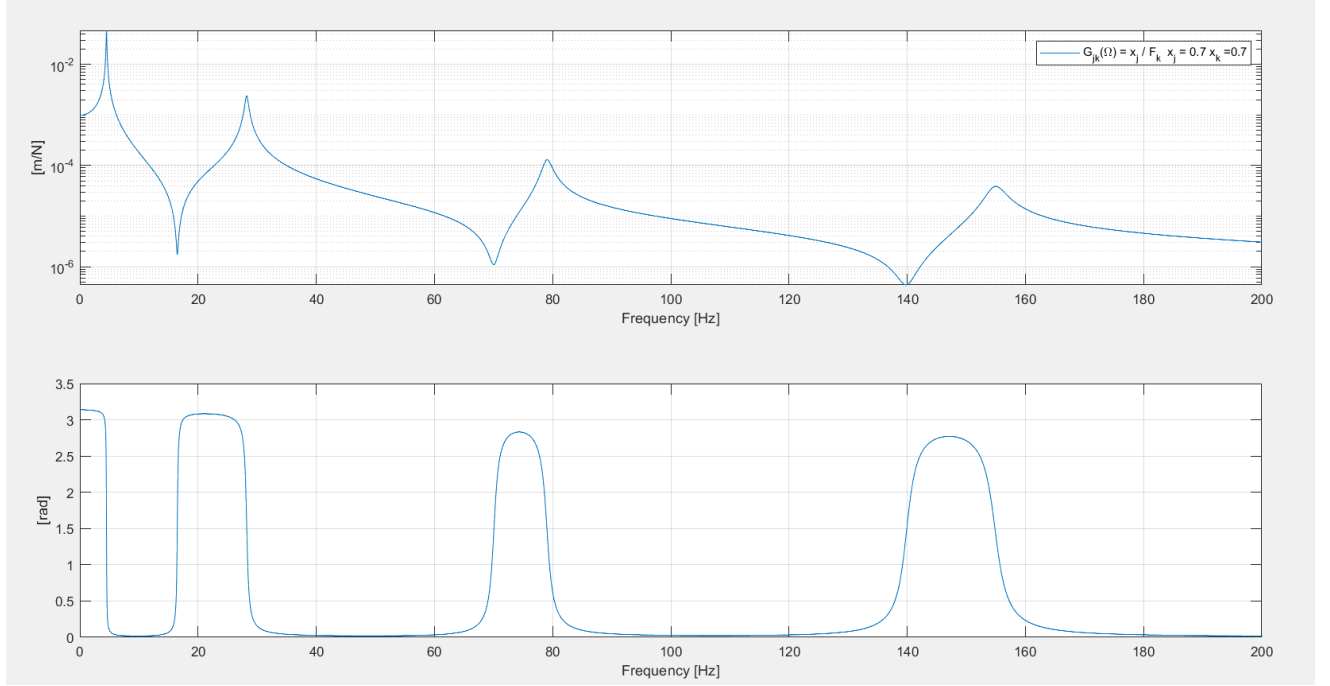
Input $x_k = 0.604$ m (node of the 3rd mode shape)



4th case: collocated FRF

Output $x_j = 0.7$ m

Input $x_k = 0.7$ m



Modal parameters identification

The modal parameters were identified through the FRF-based multi-mode curve fitting method. At first, the simplified identification methods were used to provide first attempt values for the parameters. The values of the natural frequencies ω_{oi} were already identified in the previous points, while the mode shapes $A_{jk}^{(i)}$ and the damping ratios ξ_i were obtained through the following equations.

$$A_{jk}^{(i)} = \frac{X_j^{(i)} X_k^{(i)}}{m_i} \quad \xi_i = -\frac{1}{\omega_i \cdot \left. \frac{\partial \Phi_{jk}}{\partial \Omega} \right|_{\Omega=\omega_i}}$$

The residual contributions R_{jk}^L and R_{jk}^H were assumed equal to zero.

Then, the numerical FRF was computed using the following formula.

$$G_{jk}^{NUM}(\Omega) = \sum_{i=1}^n \frac{A_{jk}^{(i)}}{-\Omega^2 + j2\xi_i\omega_i\Omega + \omega_i^2} + \frac{R_{jk}^L}{\Omega^2} + R_{jk}^H$$

The numerical FRF was compared with the experimental FRF, minimizing the error among them and applying a least square minimization.

Here is the matrix containing the obtained modal parameters.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	28.2975	0.0100	-0.1741	-0.9536	-1.6983	-2.9743	7.8055e-09	2.6290e-08	2.7539e-08	-7.2075e-09	5.4937e-06	1.8503e-05	1.9372e-05	-5.0692e-06
2	177.3425	0.0100	0.8682	2.6948	2.3959	-0.8346	-9.6393e-13	1.6779e-10	4.8033e-10	8.1460e-10	-3.9180e-08	5.2242e-06	1.4998e-05	2.5431e-05
3	496.5625	0.0100	-1.8864	-1.7530	1.5674	0.8394	3.1959e-14	-7.2689e-13	-4.3796e-13	1.2506e-12	7.7796e-09	-1.7922e-07	-1.0800e-07	3.0798e-07
4	973.0675	0.0100	2.6710	-1.6442	-1.6773	2.0542	0	0	0	0	0	0	0	0

ω_{oi}

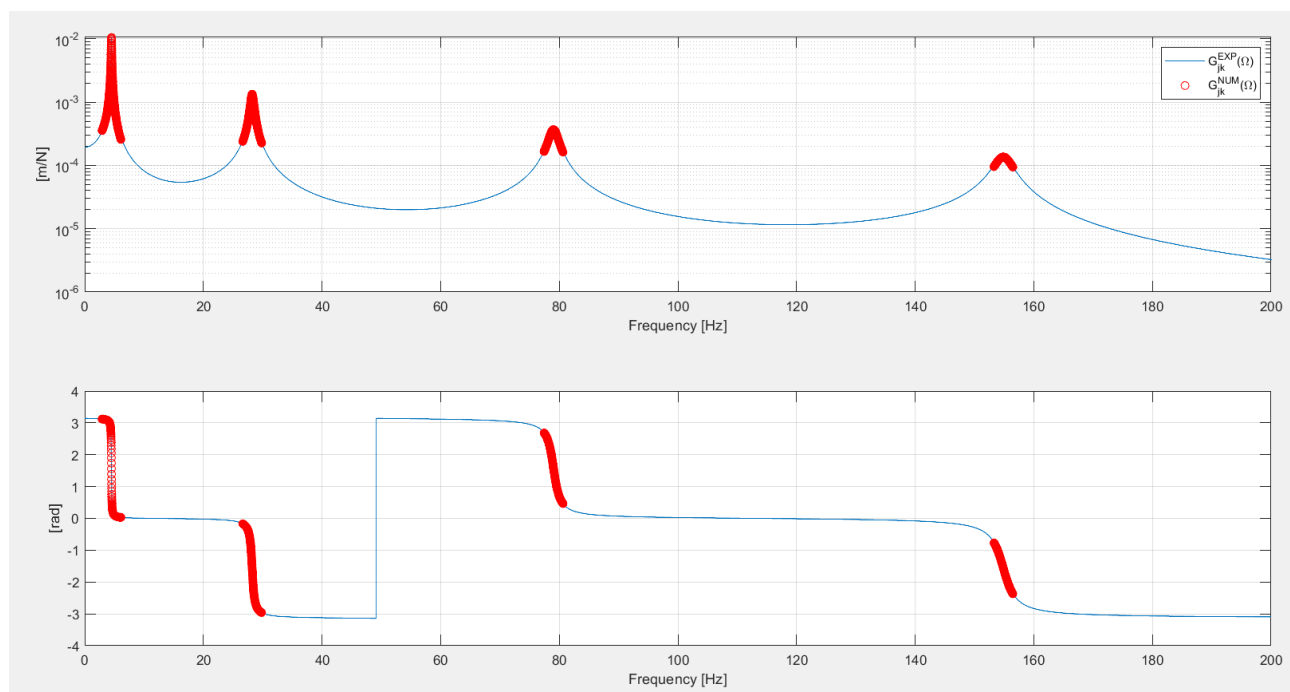
ξ_i

$A_{jk}^{(i)}$

R_{jk}^L

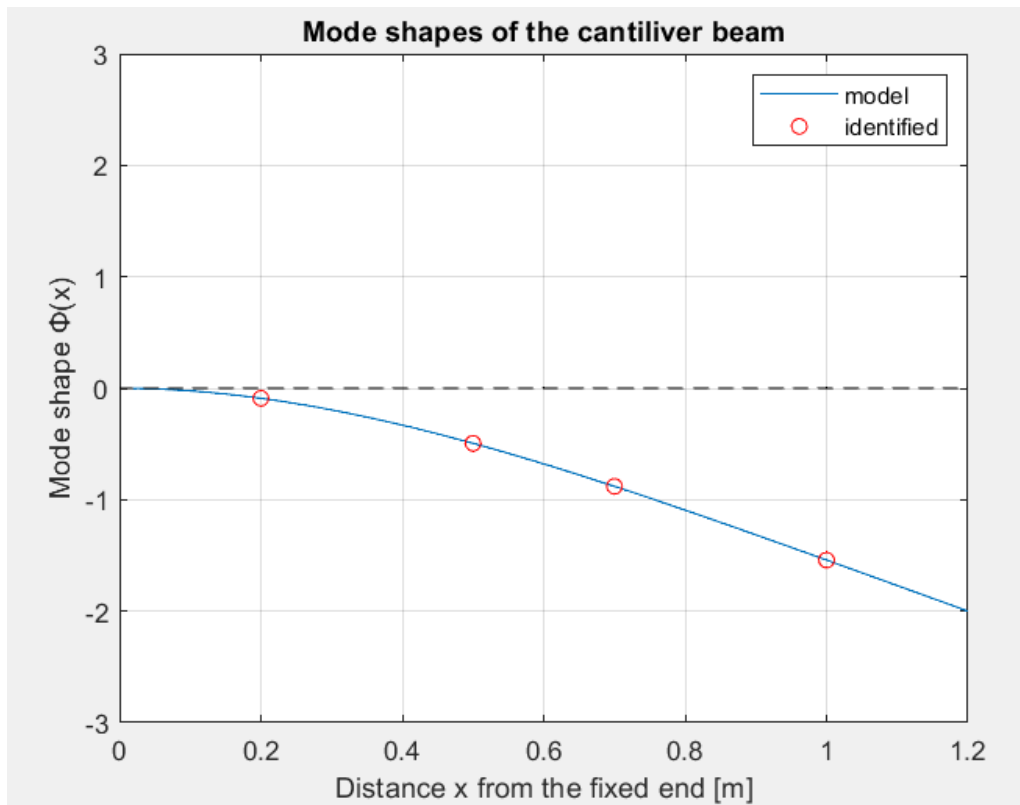
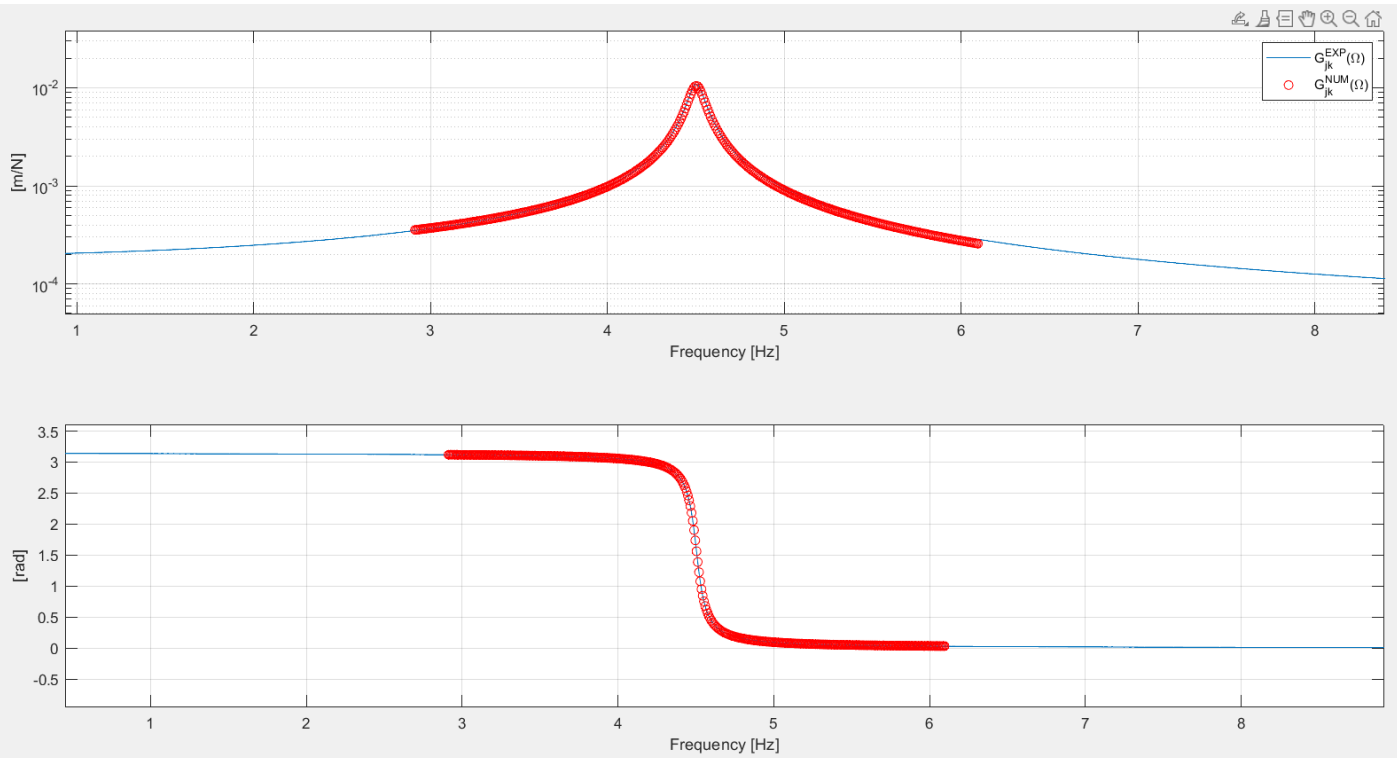
R_{jk}^H

The following pictures show the comparison of the FRFs in correspondence of the resonance regions and the parameters (identified and simulated) of each mode shape.

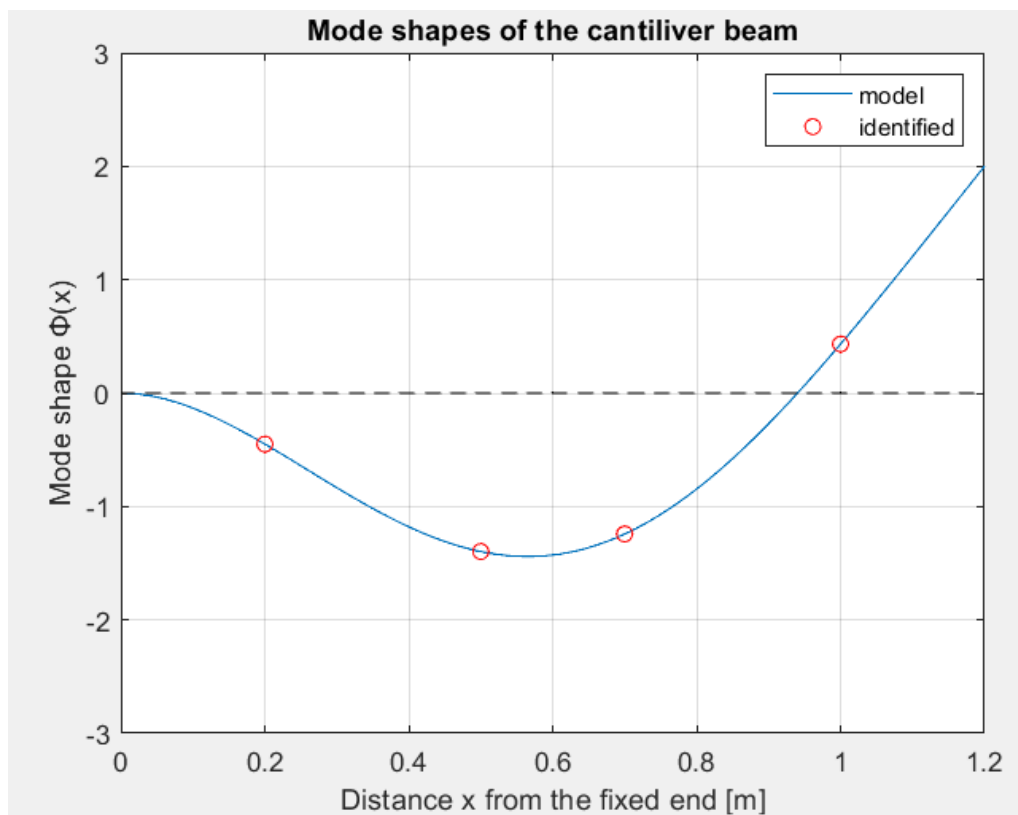
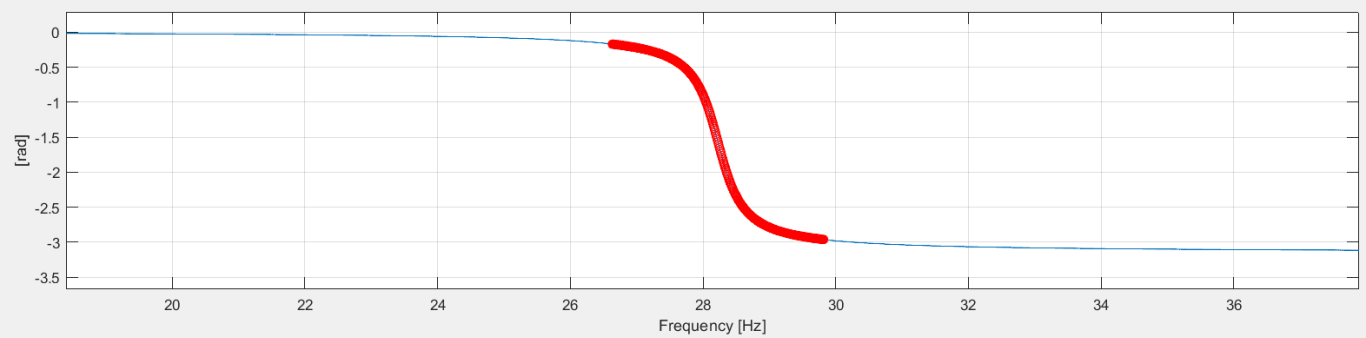
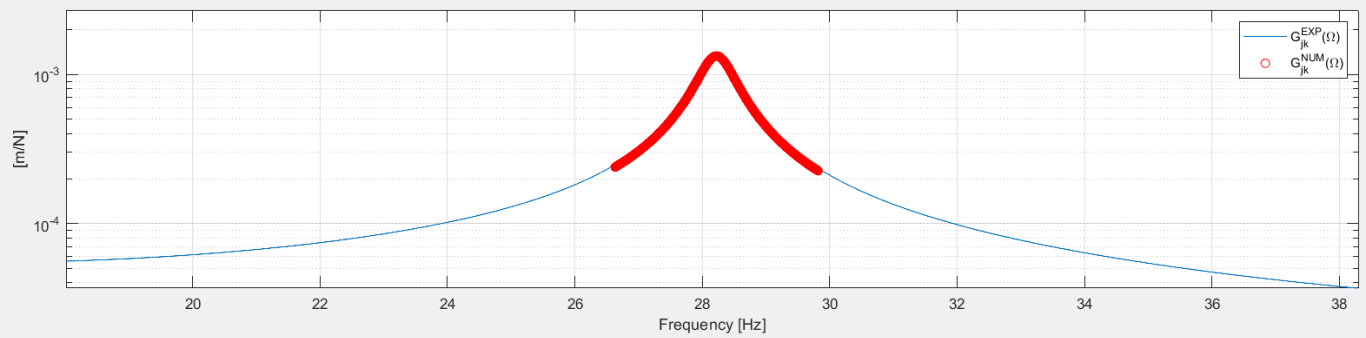


Output $x_j = 0.2m$ Input $x_k = 1.2m$

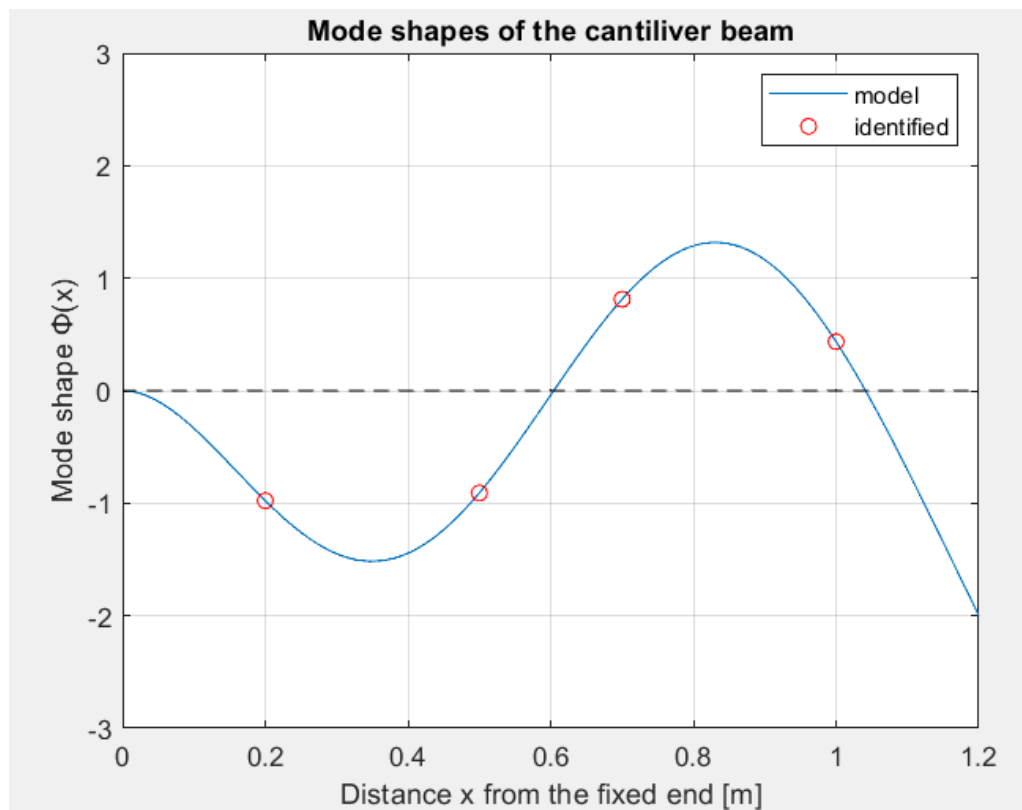
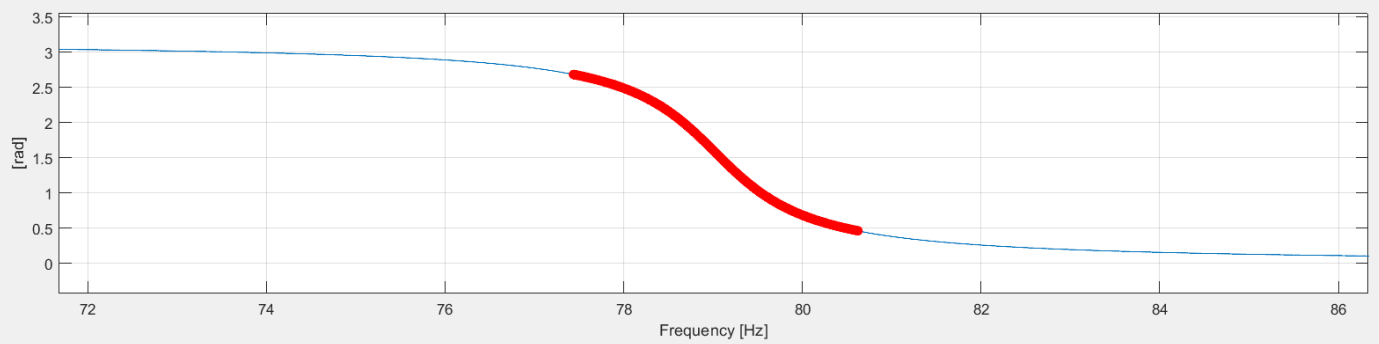
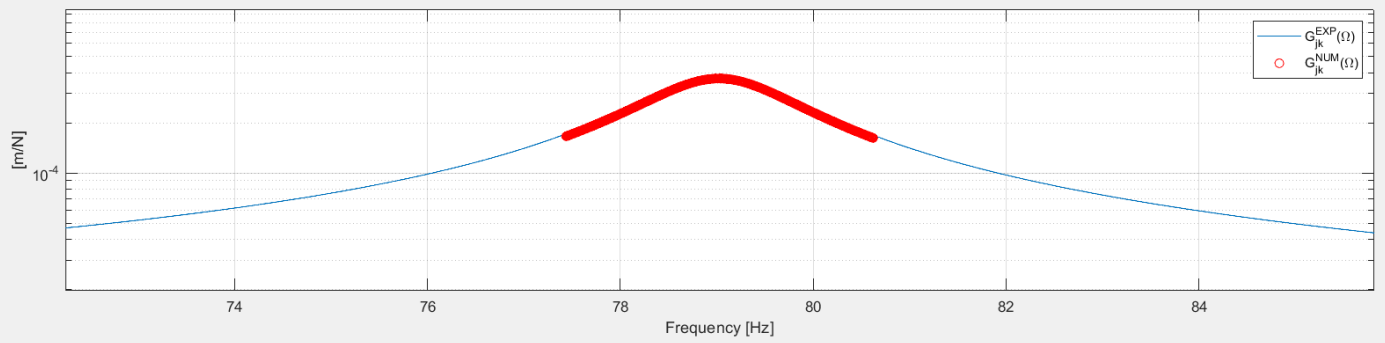
Mode 1



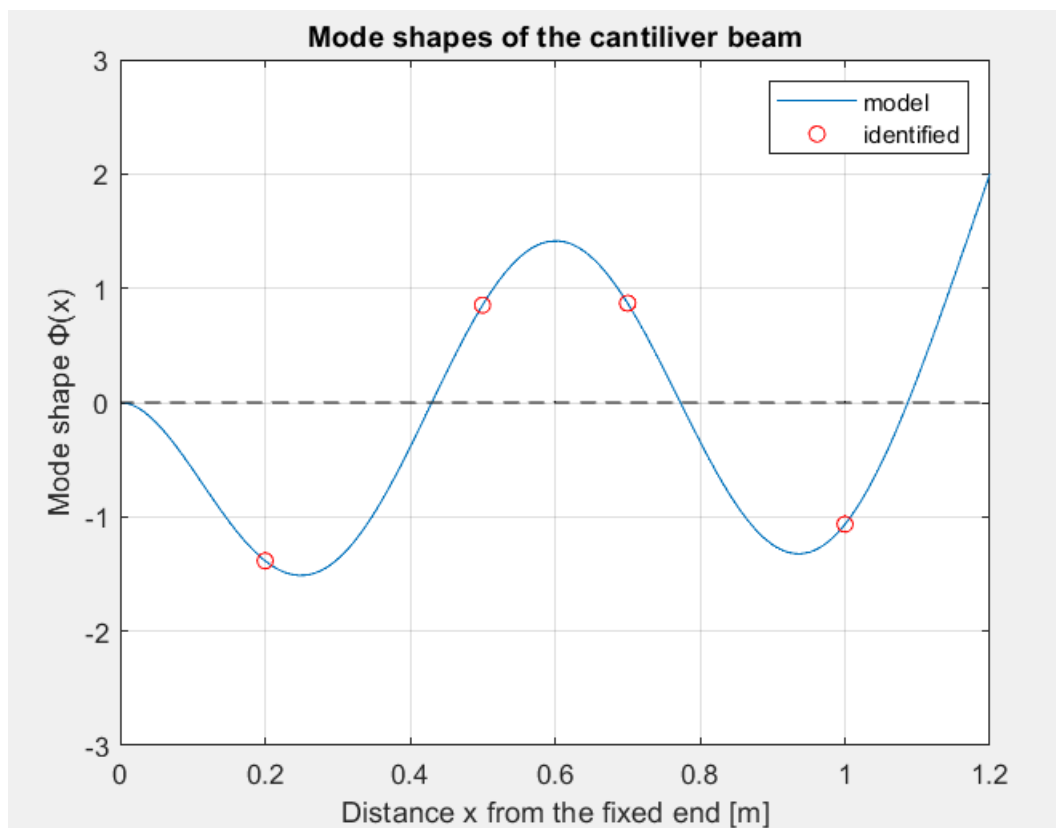
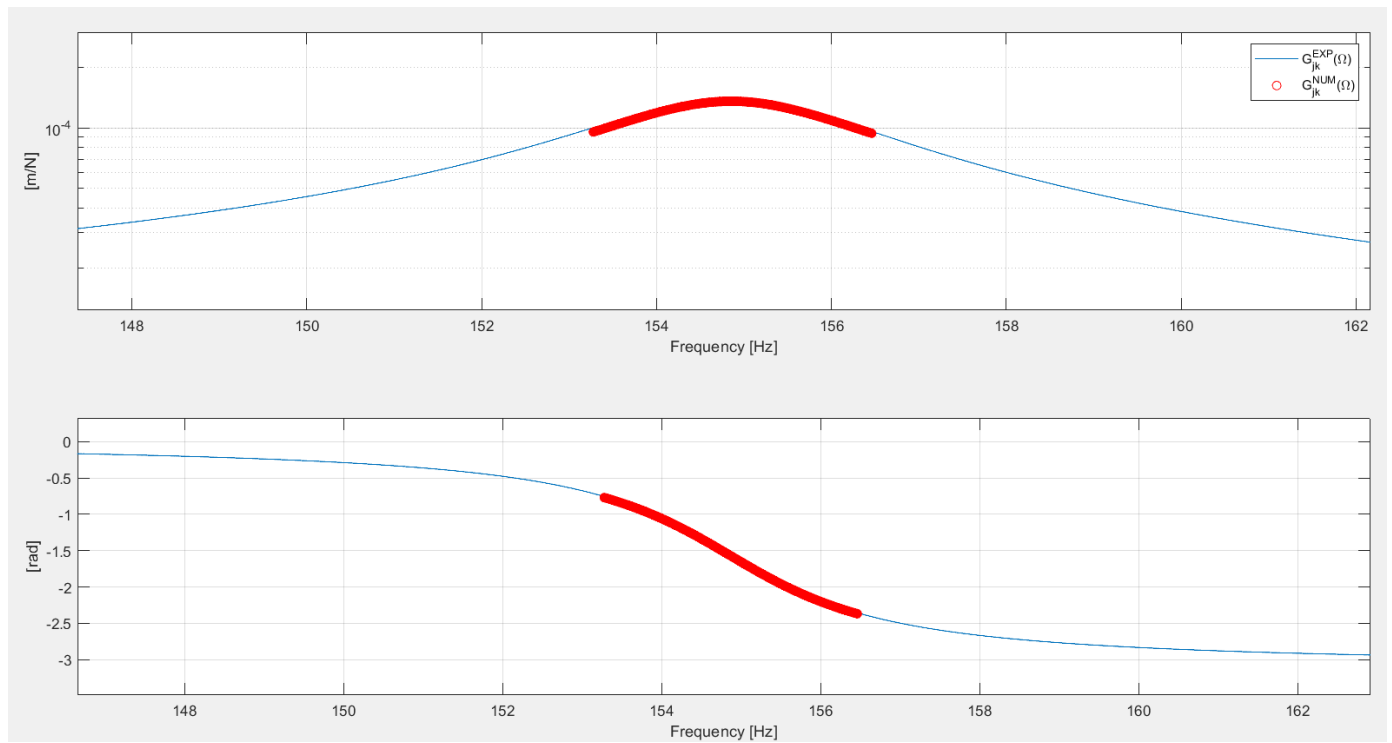
Mode 2



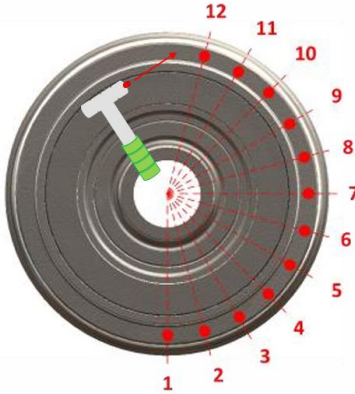
Mode 3



Mode 4

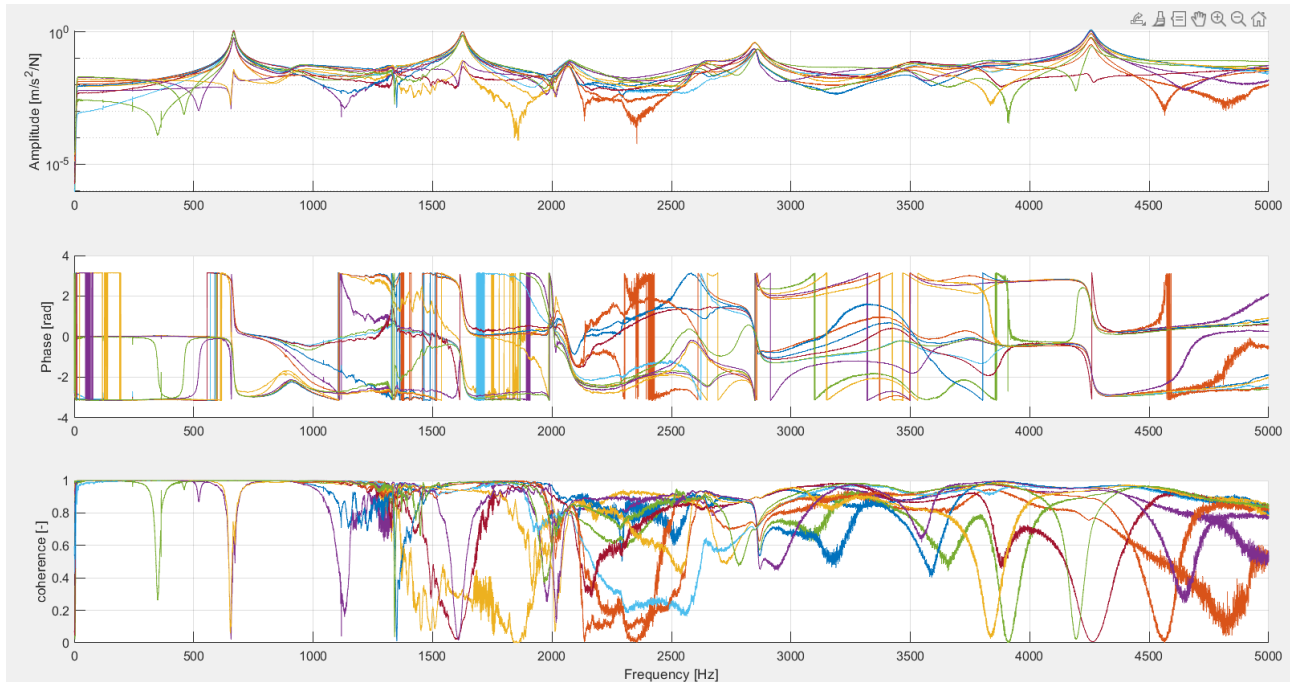


PART B: Light Rail wheel



A set of 12 axial accelerometers is placed on one half of the wheel, with a space distance between the sensors of 15 degrees each. The symmetry of the structure allows us to avoid the use of sensors on the second half of the wheel.

The Data set provided contains the 12 FRFs (and the relative coherence functions) corresponding to the response of each single sensor to the impact test performed on the wheel.



Single-mode identification procedure:

- Natural frequencies: identified from the position of the resonance peak in one of the experimental FRFs. (we select an FRF among the 12 available whose peak is evident and clearly visible)

Natural frequencies:

$$f_{01} = 667.3244 \text{ Hz} \quad f_{02} = 1625.3117 \text{ Hz}$$

- Damping ratios: identified from the slope of the phase diagram.

$$\xi_i = \frac{-1}{\omega_i \cdot \left. \frac{\partial FRF_{jk}}{\partial \Omega} \right|_{\Omega=\omega_i}} \Rightarrow \xi_1 \cong 0.008047551, \quad \xi_2 \cong 0.005981112$$

- **Mode shapes:** identified from the imaginary part of the experimental FRF (receptance or inertance), evaluated for variable output node j , in correspondence with $\Omega = \omega_i$.

$$G_{jk}(w_i) \cong \frac{x_j^{(i)} x_k^{(i)}}{j \omega_i c_i} = j \left(-\frac{x_k^{(i)}}{\omega_i c_i} \right) x_j^{(i)} = j \left(-\frac{x_k^{(i)}}{2 m_i \xi_j \omega_i^2} \right) x_j^{(i)}$$

$$A_{jk}^{(i)} = \frac{X_k^{(i)} x_j^{(i)}}{m_i} \Rightarrow A_{jk}^{(i)} = -j (G_{jk}(w_i) \cdot 2 \cdot \xi_i \cdot w_i^2)$$

- $R_{jk}^{Lower\ freq}, R_{jk}^{Higher\ freq} = 0$ for every j

Results from the least-square minimization:

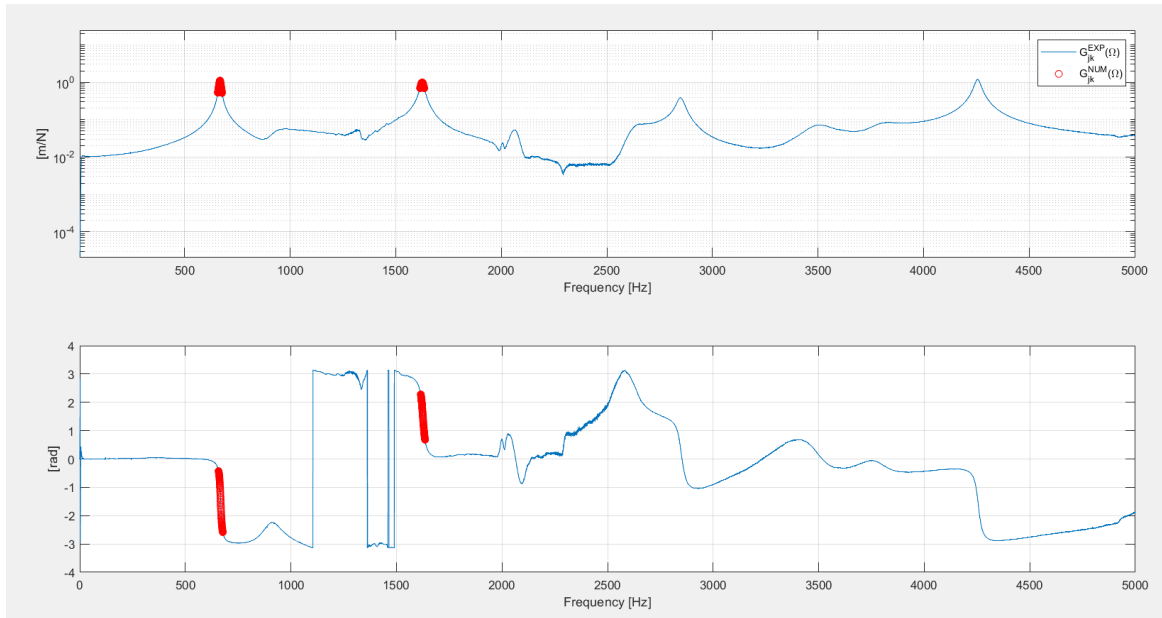
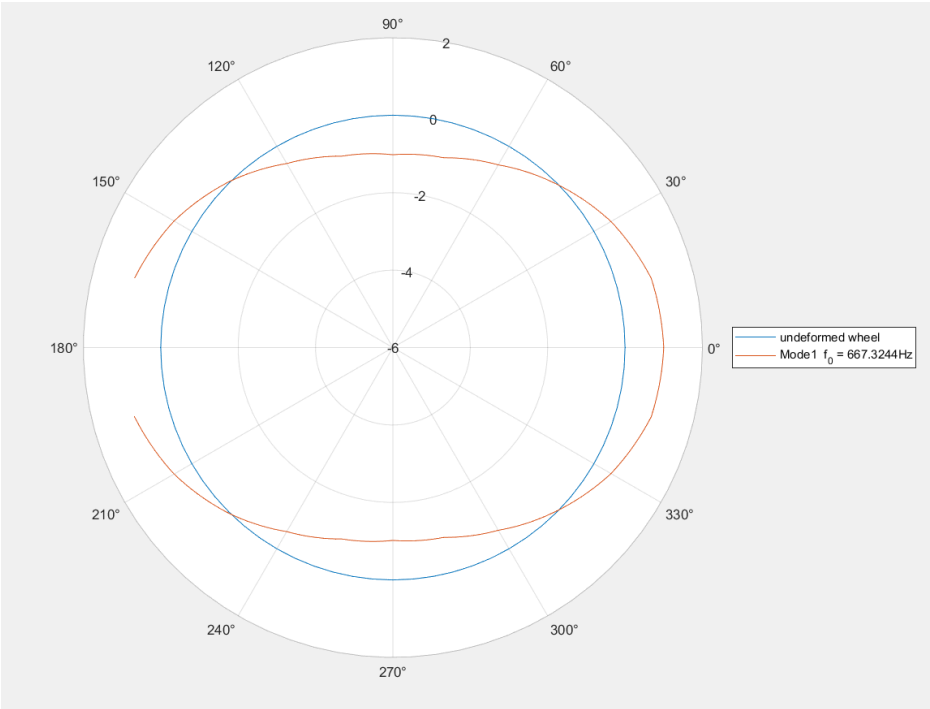
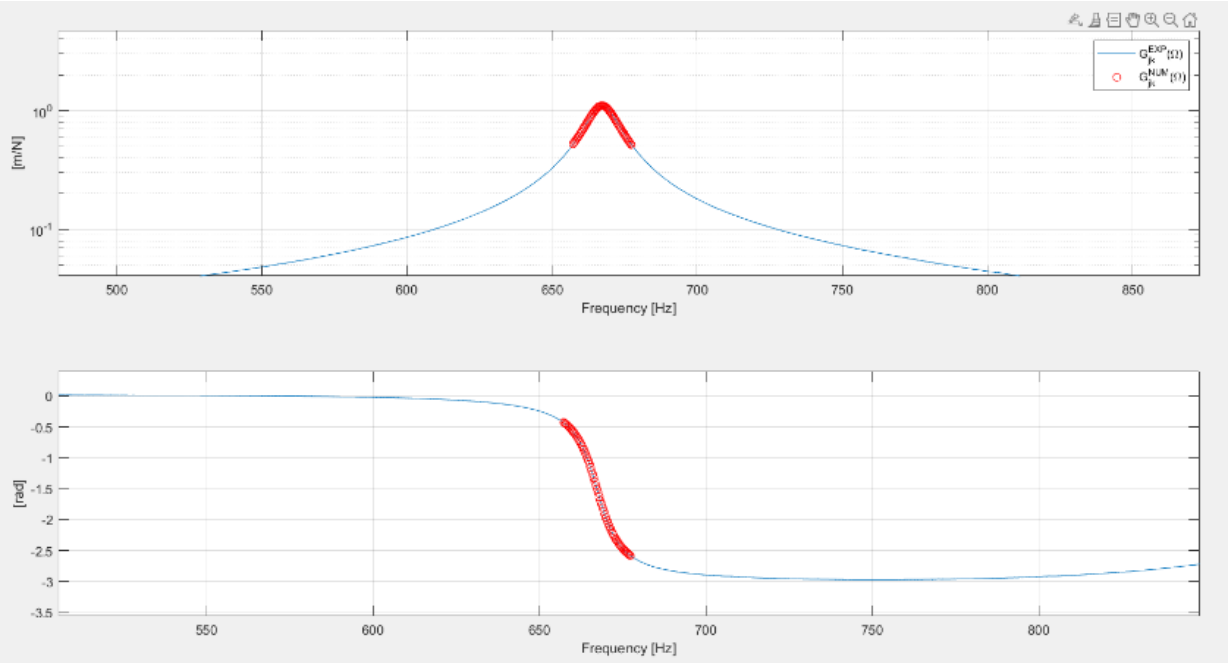
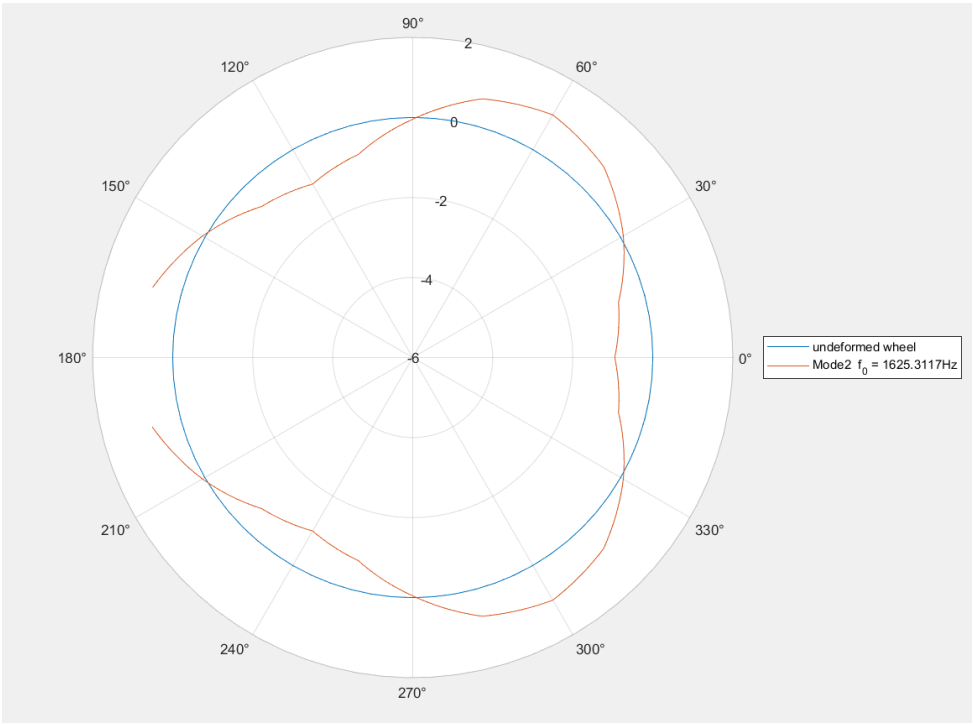
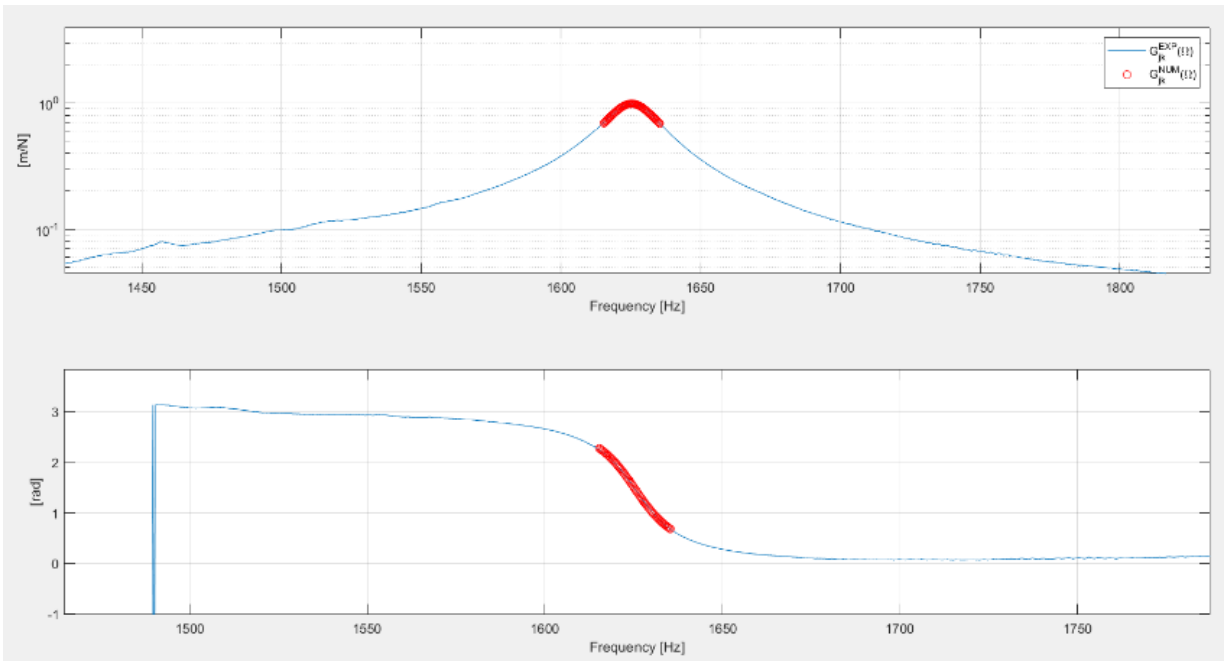


Figure 3: Result of the identification (on the 1st experimental FRF of the data set)

Mode 1



Mode 2



Comments:

- In order to have a smooth and continue mode shape representations, a linear interpolation has been performed between each measurement point:

```
Modeshapes_scaled= Modeshapes ./ max(Modeshapes);  
theta_rad = deg2rad(0:15:165);  
theta_inter = deg2rad(linspace(0,165,200));
```

```
modeshape_interp= interp1(theta_rad,Modeshapes_scaled,theta_inter);
```

- The arrangement of the accelerometers along the wheel does not allow a complete representation of its mode shapes. Having no sensor located at the point where the impact test is carried out (collocated FRF), we'll have a lack of informations regarding the behavior of the wheel in that particular position.

Considering the symmetry of the wheel and a separation of 15 degrees between one sensor and another, the representation of the mode shapes covers only 330 degrees.

Assignment 2

In-plane Finite Element Model of a Truss Bridge

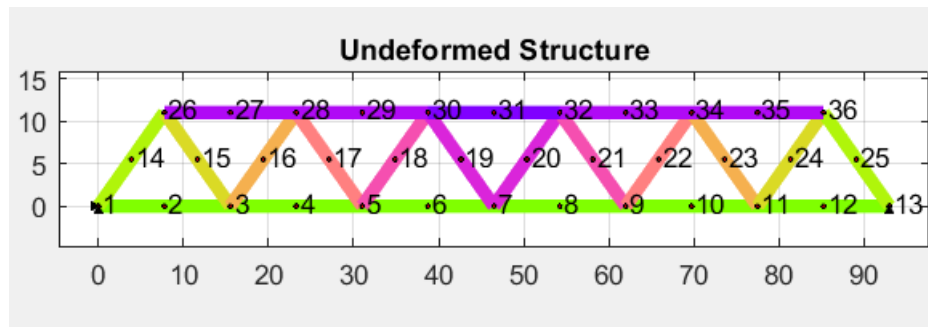
FE Model:

Frequency range: 0 – 7 Hz:

$$\Omega_{max} = 7 \cdot 2\pi \frac{rad}{s}$$

$$\omega_k^{(1)} = \left(\frac{\pi}{L_k}\right)^2 \sqrt{\frac{EJ_k}{m_k}} > 1.5\Omega_{max} \Rightarrow L_{max,k} = \sqrt{\frac{\pi^2}{1.5\Omega_{max}} \sqrt{\frac{EJ_k}{m_k}}}$$

Beam element	Deck element	Diag member K1	Diag member K2	Diag member K3	Diag member K4	Diag member K5	Diag member K6	Upper chord G1	Upper chord G2
$L_{max} [m]$	10.6567	10.9119	10.4422	10.5287	8.7001	9.8625	7.8886	15.0064	15.0517



Structure properties:

N° beam = 46

$L = 7.7500m$ (horizontal beams), $L = 6.7280m$ (oblique beams)

N° nodes = 36

N° degrees of freedom = 105 (+ 3 constrained = x1, y1, y13)

Natural frequencies and mode shapes:

```
[M,K]=assem(incid,l,m,EA,EJ,gamma,idb);
MFF = M(1:ndof, 1:ndof);
KFF = K(1:ndof, 1:ndof);

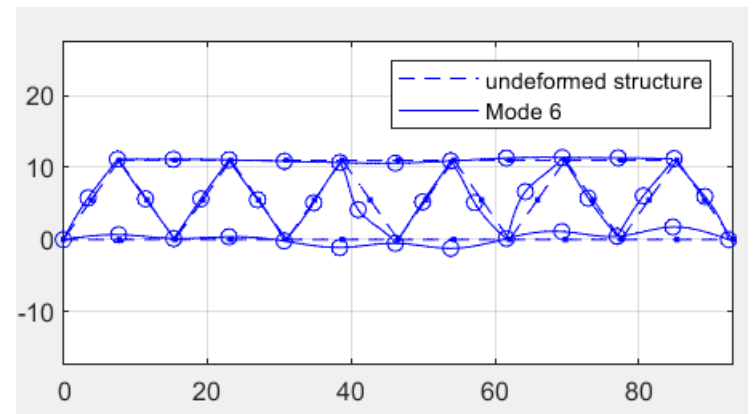
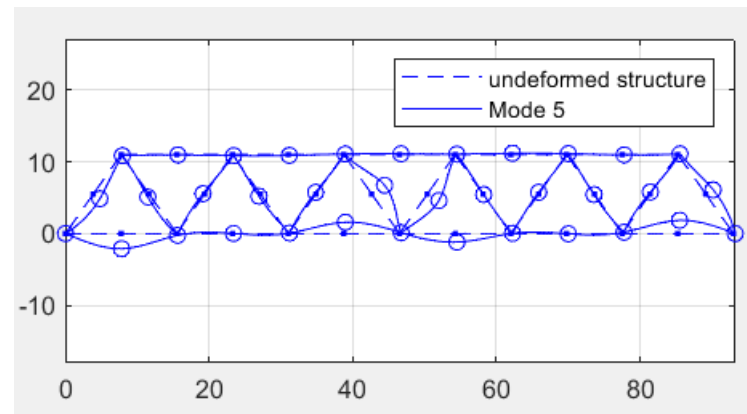
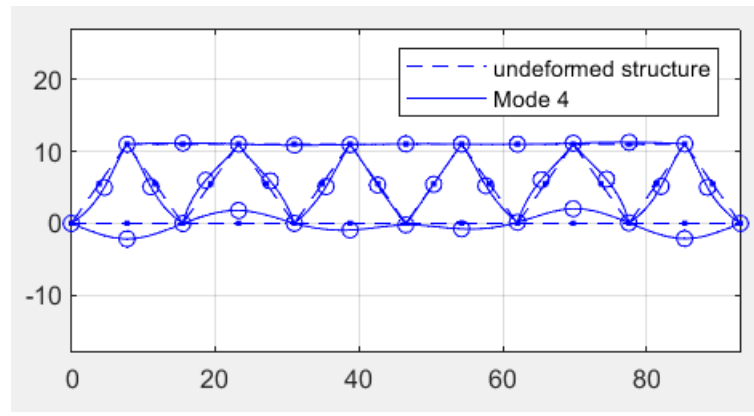
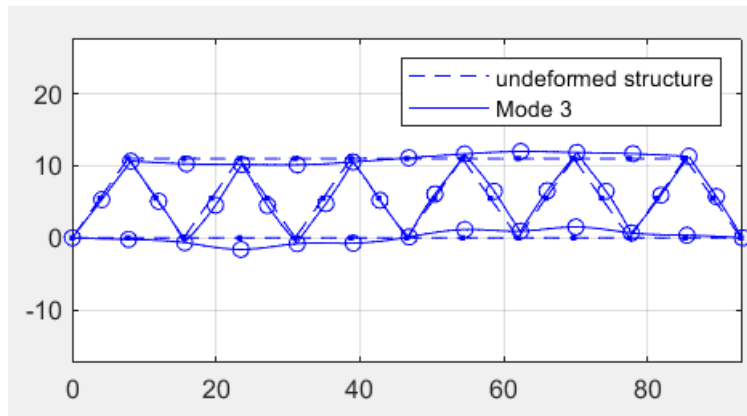
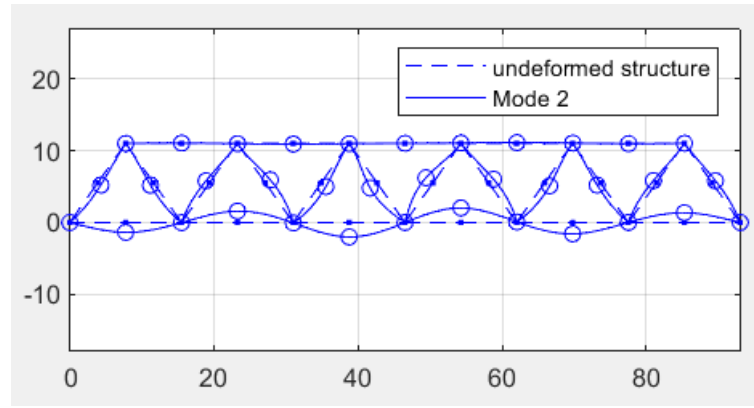
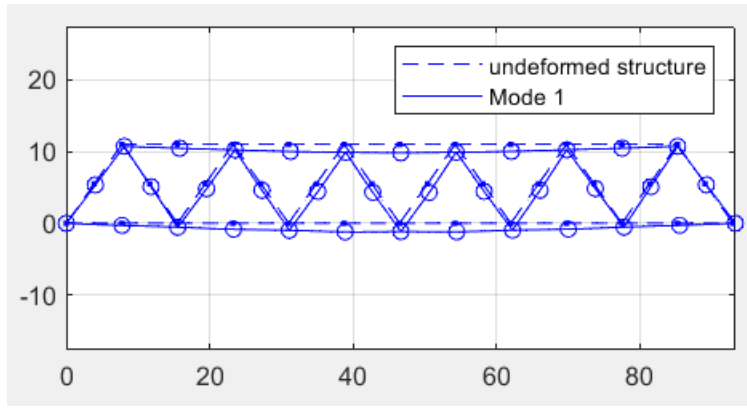
[modes, omega2] = eig(inv(MFF)*KFF);
omega = diag(sqrt(omega2));
```

$$[M] = \begin{bmatrix} [M_{FF}] & [M_{FC}] \\ [M_{CF}] & [M_{CC}] \end{bmatrix}$$

$$[K] = \begin{bmatrix} [K_{FF}] & [K_{FC}] \\ [K_{CF}] & [K_{CC}] \end{bmatrix}$$

1	2	3	4	5	6
15.8190	32.9057	34.9432	35.6091	40.1835	43.1370

Natural frequencies of the structure up to the 6th (rad/s)



Damping matrix:

Rayleigh damping: $[C] = a[M] + \beta[k]$

$$c_i = am_i + \beta k_i \Rightarrow \xi_i = \frac{c_i}{2m_i\omega_i} = \frac{a}{2\omega_i} + \frac{\beta\omega_i}{2}$$

By imposing $\xi_1 = 1\%$, $\xi_2 = 0.75\%$ we can find the values of a and β by same matrix calculations:

$$\begin{bmatrix} \frac{1}{2\omega_1} & \frac{\omega_1}{2} \\ \frac{1}{2\omega_2} & \frac{\omega_2}{2} \end{bmatrix} \begin{bmatrix} a \\ \beta \end{bmatrix} = \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} \Rightarrow [A] \begin{bmatrix} a \\ \beta \end{bmatrix} = \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} \quad , \quad \begin{bmatrix} a \\ \beta \end{bmatrix} = [A]^{-1} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} = \begin{bmatrix} 0.2631 \\ 2.1285e - 0.4 \end{bmatrix}$$

As matrix A is a 2x2 matrix the system is determined.

FRF

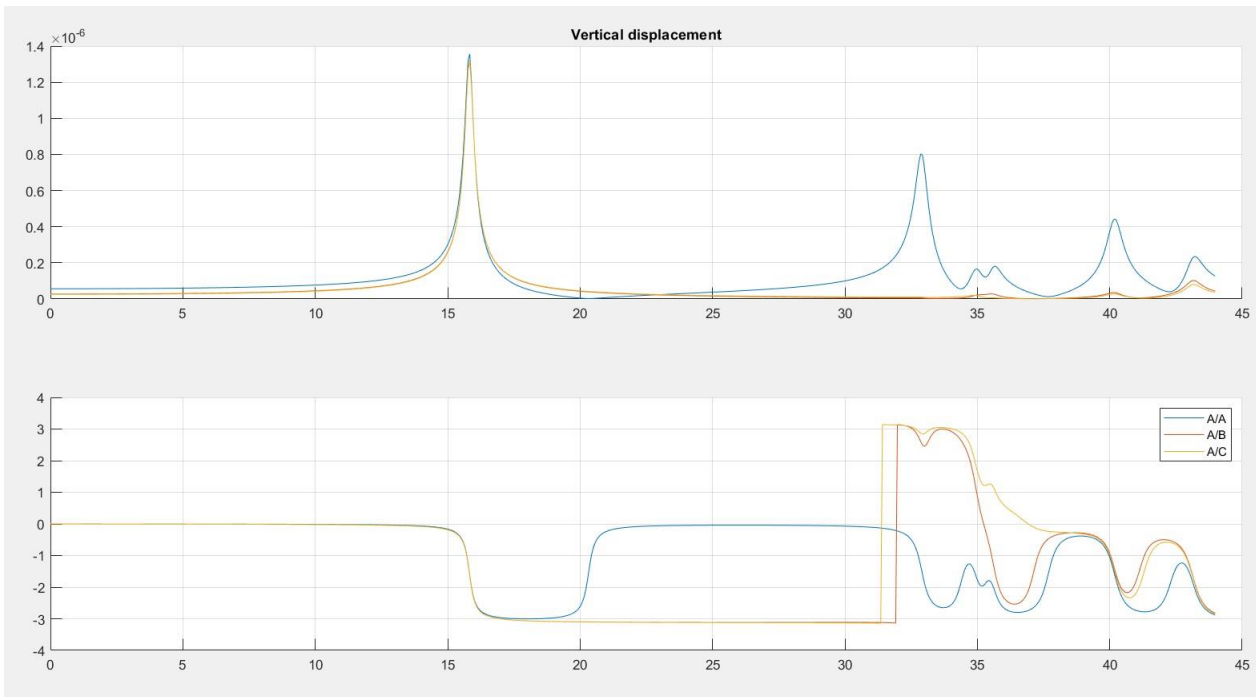
The FRF can be computed by solving:

$$-\Omega^2[M_{FF}] + j\Omega[C_{FF}] + [K_{FF}]X_0 e^{j\Omega t} = F_0 e^{j\Omega t}$$

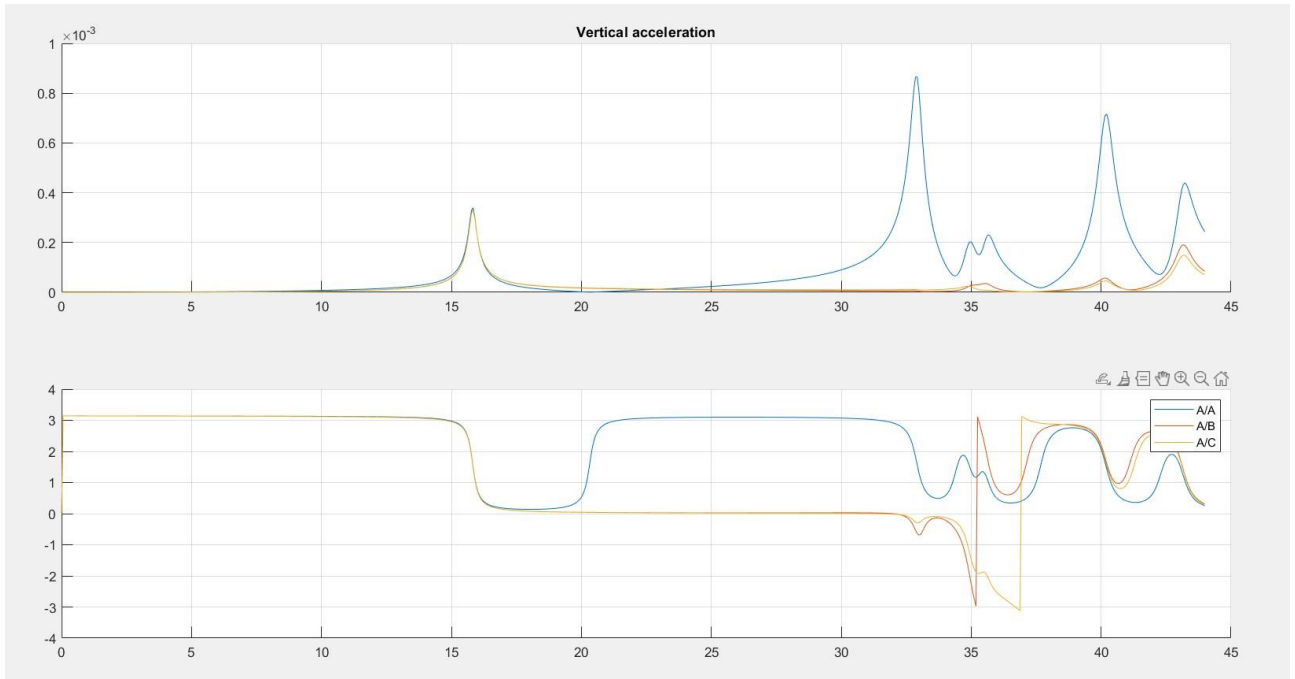
Where the external force vector is set to 1 only for the index related to the vertical dof of node A

$$F_0 = [0 \dots 0 \ 1 \ 0 \dots 0]$$

We get as many FRF as the number of dofs of the system and we plot the corresponding ones at vertical displacement of nodes A, B, C.



The FRFs for the vertical acceleration are computed differentiating two times with respect to time



The axial and the transverse displacements of a beam section are defined as functions of the nodal displacements through the shape functions

$$u(\xi, t) = a + b \xi$$

$$w(\xi, t) = a + b\xi + c\xi^2 + d\xi^3$$

The bending moment and the shear force are proportional to the second and third partial derivatives of w :

$$N = EA \frac{\partial u}{\partial \xi} \quad T = EJ \frac{\partial^3 w}{\partial \xi^3} \quad M = EJ \frac{\partial^2 w}{\partial \xi^2}$$

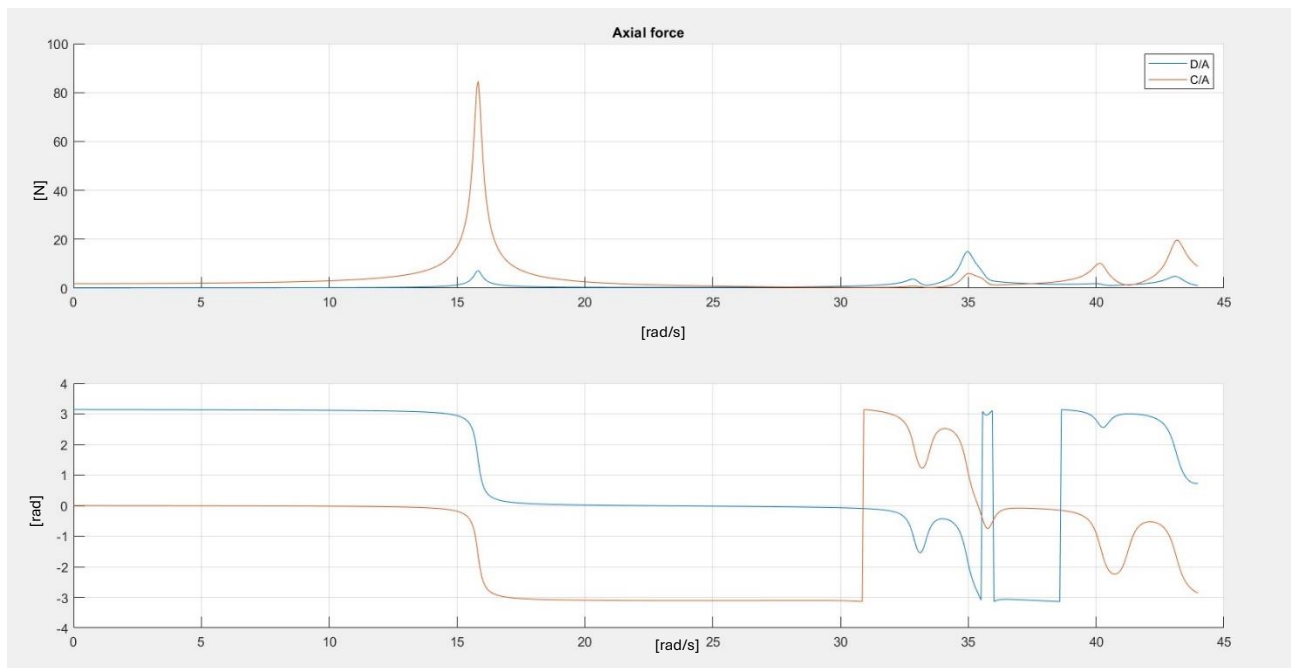
Substituting:

$$N = EA f_u' x^L(t)$$

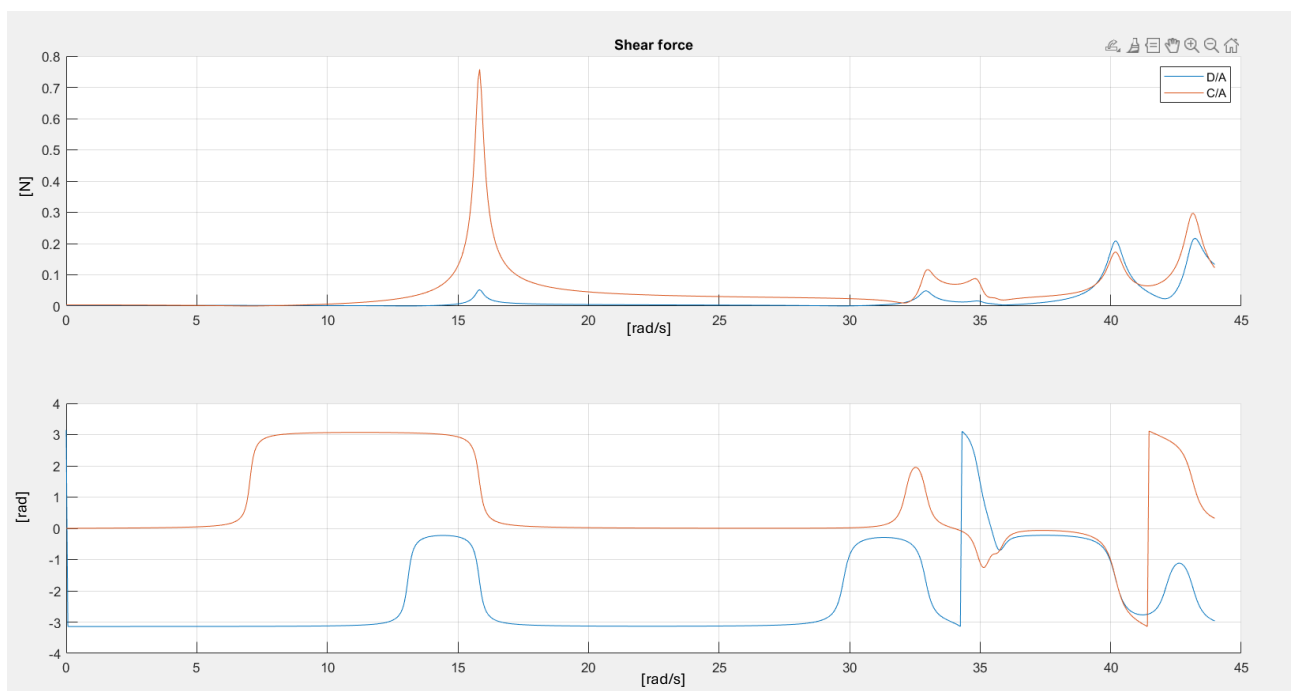
$$T(\xi) = EJ f_w'''(\xi) x^L(t)$$

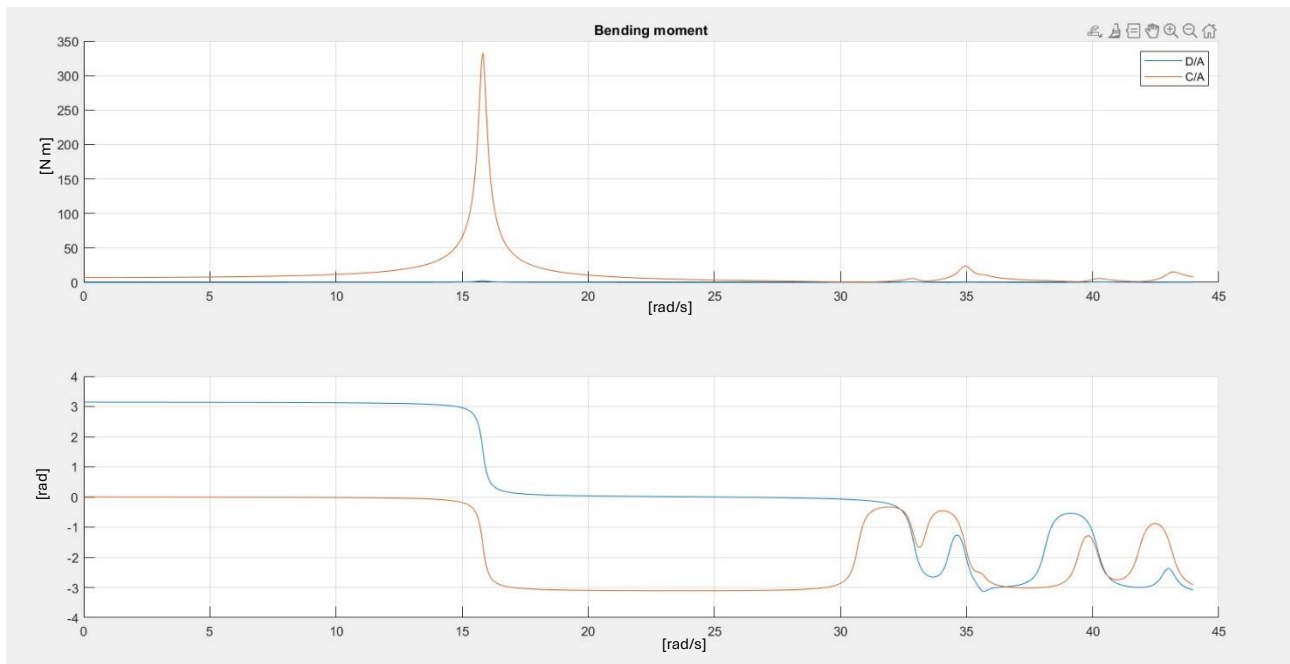
$$M(\xi) = EJ f_w''(\xi) x^L(t)$$

The magnitude and phase diagrams for nodes C and D are plotted.



We obtain





Constraint forces

Startin from the EOMs:

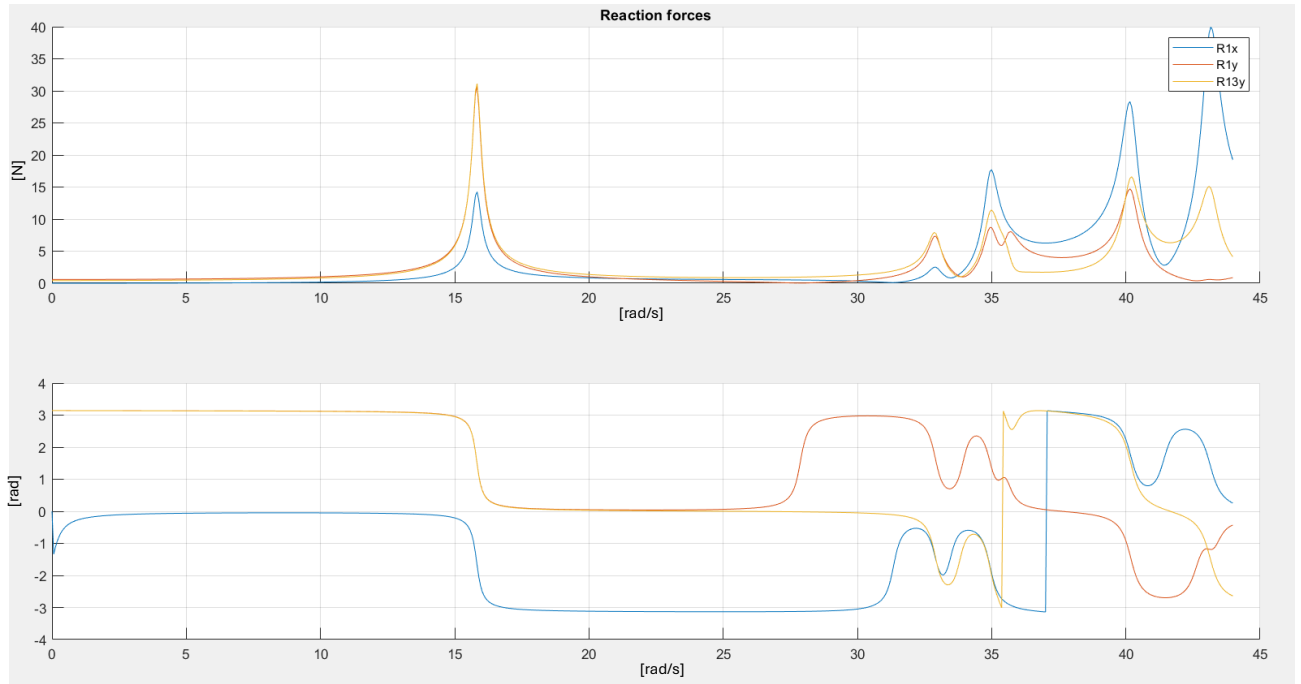
$$[M]\ddot{x} + [C]\dot{x} + [K]x = F$$

$$x_c = 0$$

$$F_c = 0$$

$$\begin{cases} [M_{FF}]\ddot{x}_F + [M_{FC}]\ddot{x}_C + [C_{FF}]\dot{x}_F + [C_{FC}]\dot{x}_C + [K_{FF}]x_F + [K_{FC}]x_C = F_F \\ [M_{CF}]\ddot{x}_F + [M_{CC}]\ddot{x}_C + [C_{CF}]\dot{x}_F + [C_{CC}]\dot{x}_C + [K_{CF}]x_F + [K_{CC}]x_C = F_C + R \end{cases}$$

$$\begin{cases} X = \left(-\Omega^2 [M_{FF}] + j\Omega [C_{FF}] + [K_{FF}] \right)^{-1} F_0 \\ R = \left(-\Omega^2 [M_{CF}] + j\Omega [C_{CF}] + [K_{CF}] \right) X \end{cases}$$



Modal superposition approach

Modal superposition approach has been applied considering the first three modes:

$$[\phi] = [X^{(1)} X^{(2)} X^{(3)}]$$

$$[M_q] = [\phi]^T [M_{FF}] [\phi]$$

$$[K_q] = [\phi]^T [K_{FF}] [\phi]$$

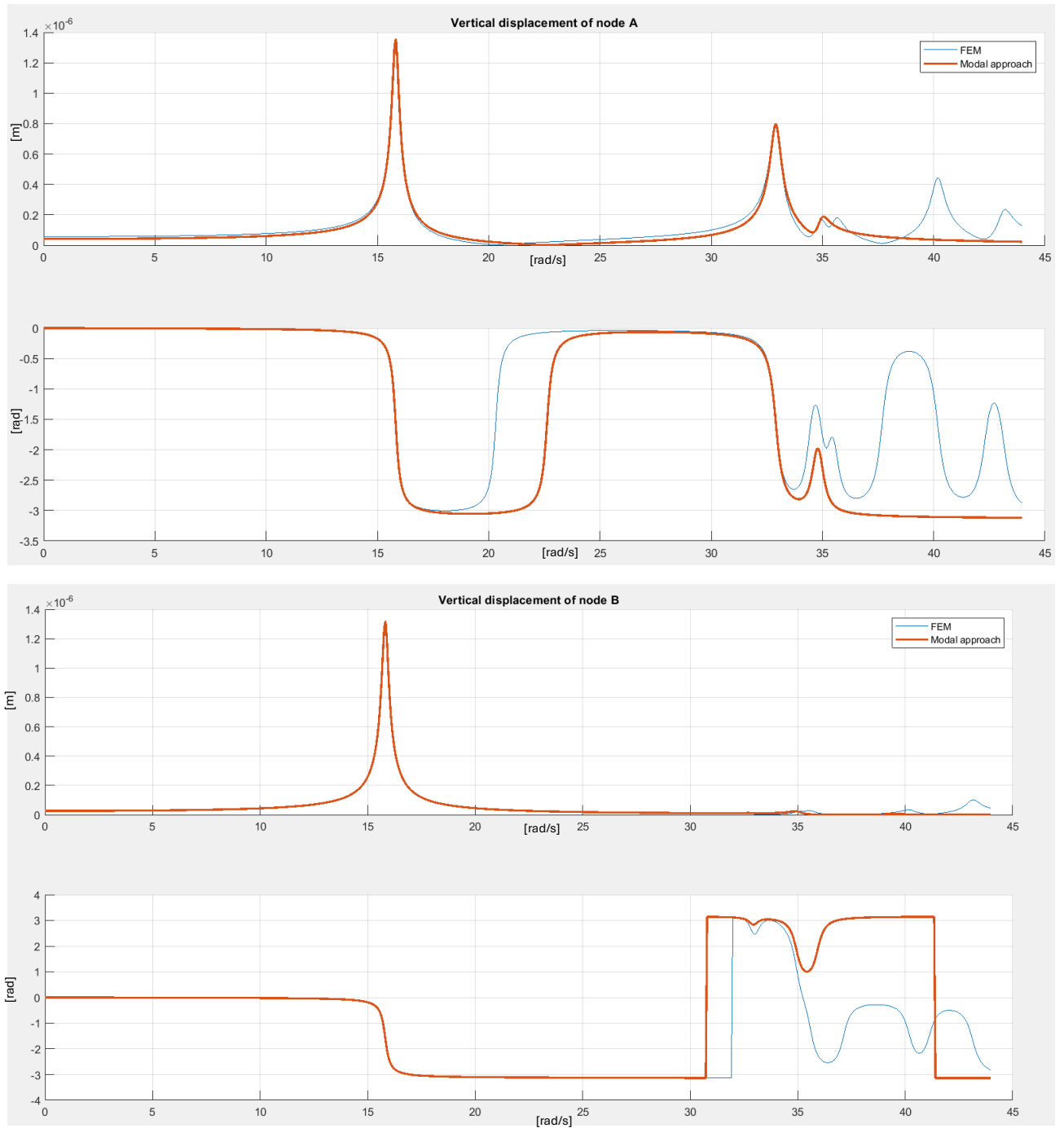
$$[C_q] = [\phi]^T [C_{FF}] [\phi]$$

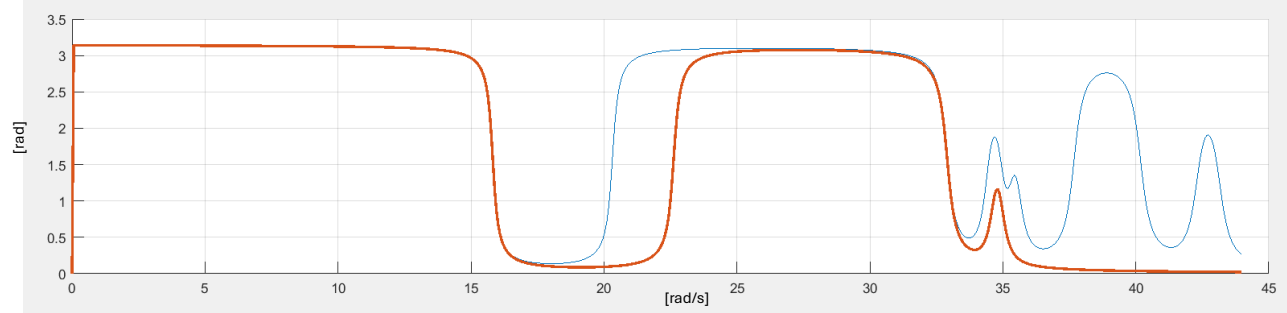
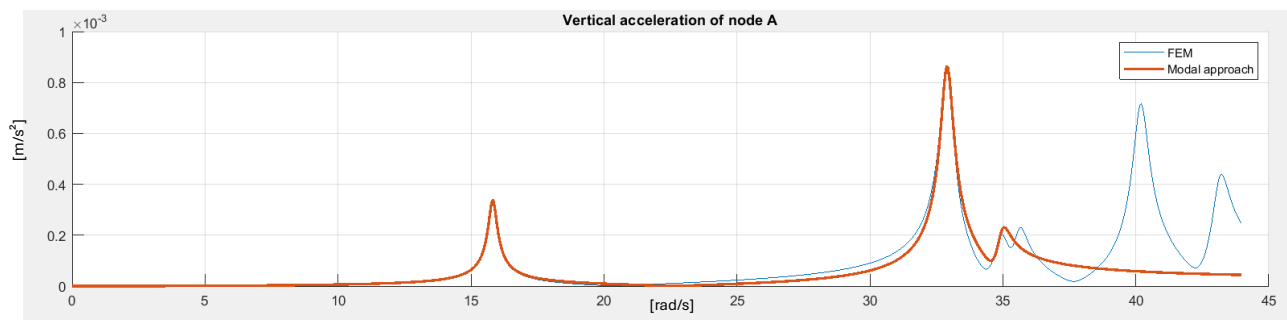
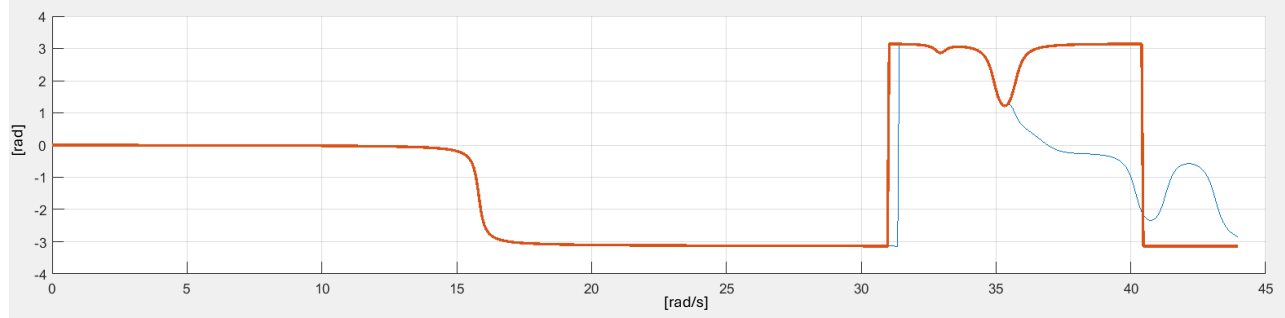
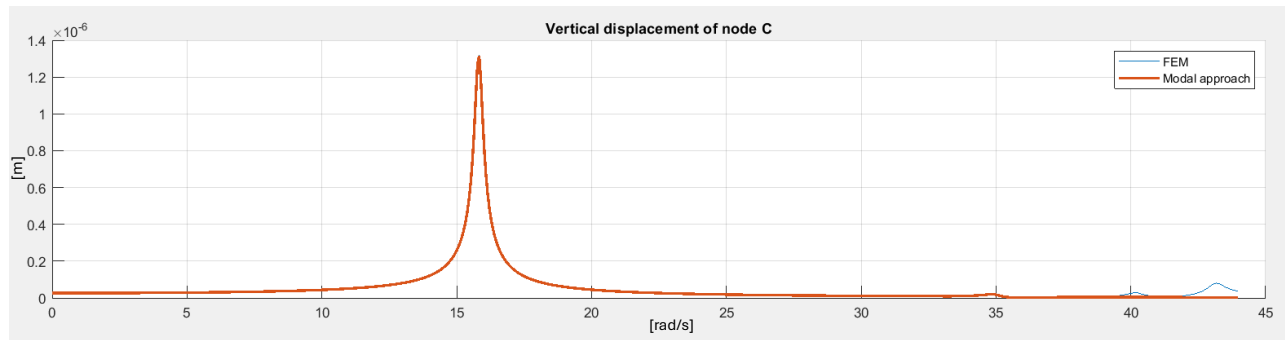
$$[Q_q] = [\phi]^T F$$

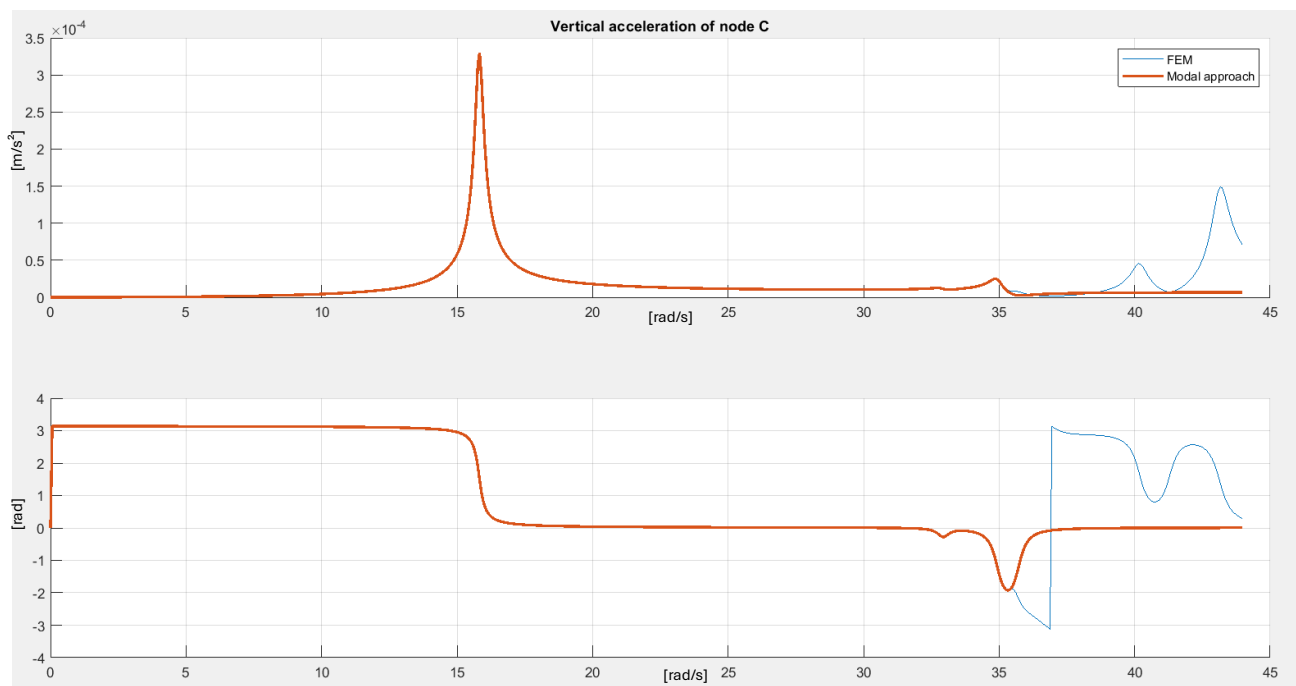
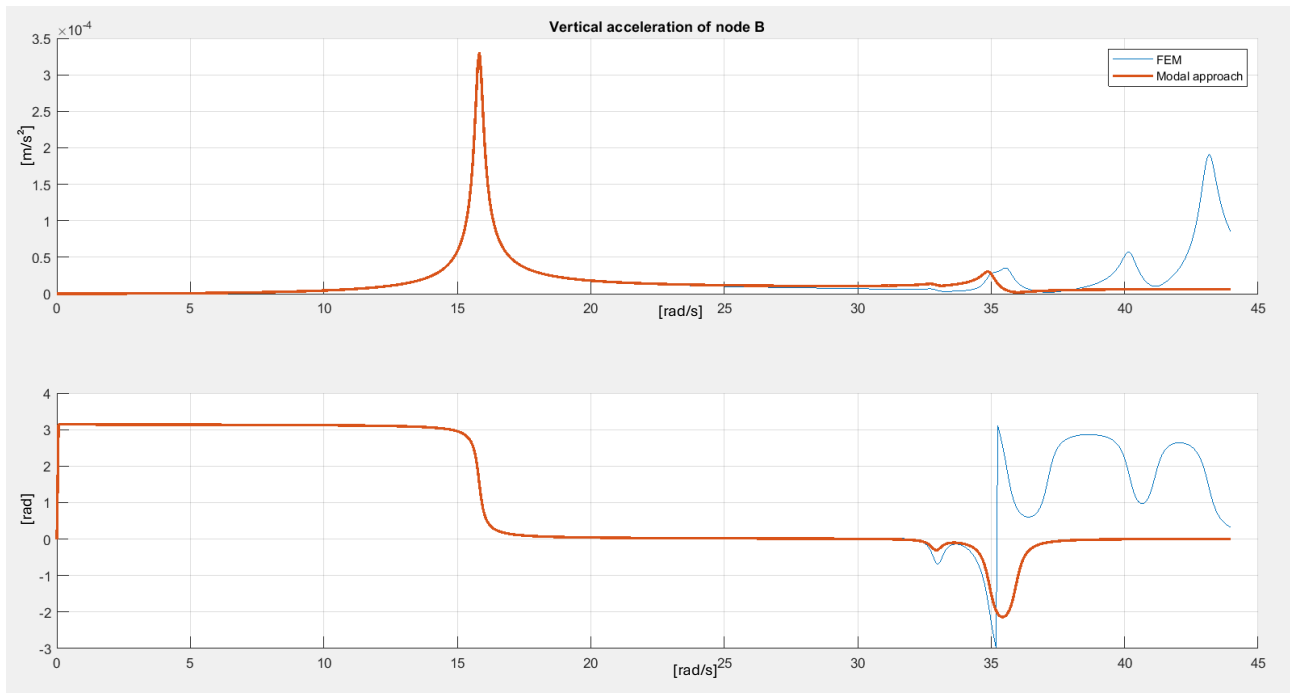
The FRFs obtained solving

$$(-\Omega^2[M_q] + j\Omega[C_q] + [K_q])q_0 e^{j\Omega t} = Q_{0q} e^{j\Omega t}$$

Has been plotted alongside the results from FEM approach







The modal superposition approach results less accurate than the FEM due to inclusion of only the first three modes, however considering a higher number of modes superposition behavior aligns closely with Fem results.

TMD

A tuned mass damper (TMD) is a device consisting of a mass m_2 , a spring k_2 and a damper c_2 . It is mounted on structures in order to reduce the vibration amplitude of a specific mode. Its natural frequency is “tuned” with the resonance peak of the mode of interest, in our case the first one. When it is excited by an external force, the damper will resonate out of phase with the structure dissipating energy. The goal is to reduce by at least 15% the amplitude of the vertical vibration of the node A. The other requirements deal with the mass and the damping ratio:

- The increment of mass of the entire structure must not exceed 2% :

$$m_2 < 0.02 m_{tot}$$

Where:

$$m_{tot} = \sum_i m_i * l_i.$$

- The damping ratio of the TMD alone must not exceed 30%.

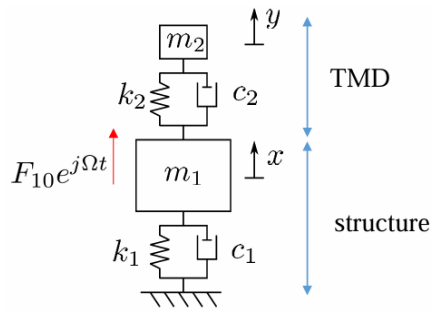
To analyze the effect of the TMD, the modal approach has been used only on the dof of interest.

$$m_1 = \phi^T M_{FF} \phi$$

$$c_1 = \phi^T C_{FF} \phi$$

$$k_1 = \phi^T K_{FF} \phi$$

$$F_1 = \phi * F_0$$



Resulting in:

$$m_1 = 8\,374.79 \text{ Kg}$$

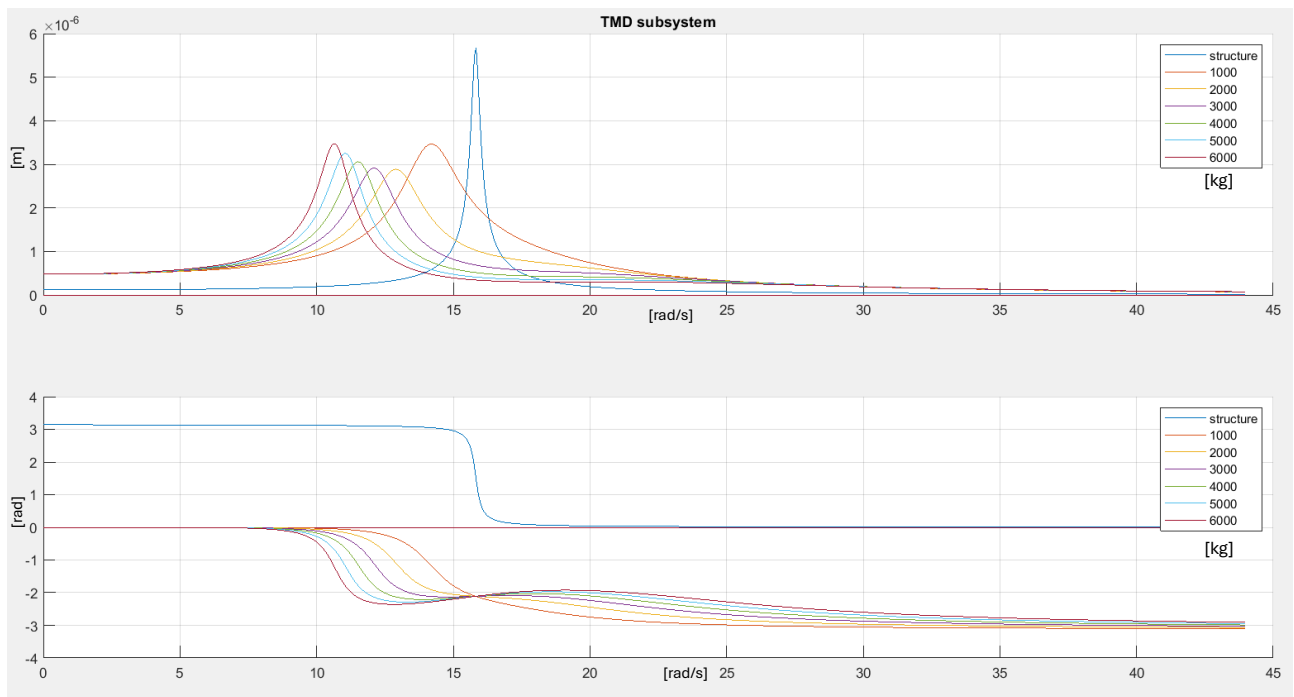
$$c_1 = 2.095 * 10^6 \frac{\text{Ns}}{\text{m}}$$

$$f_1 = -0.239 \text{ N}$$

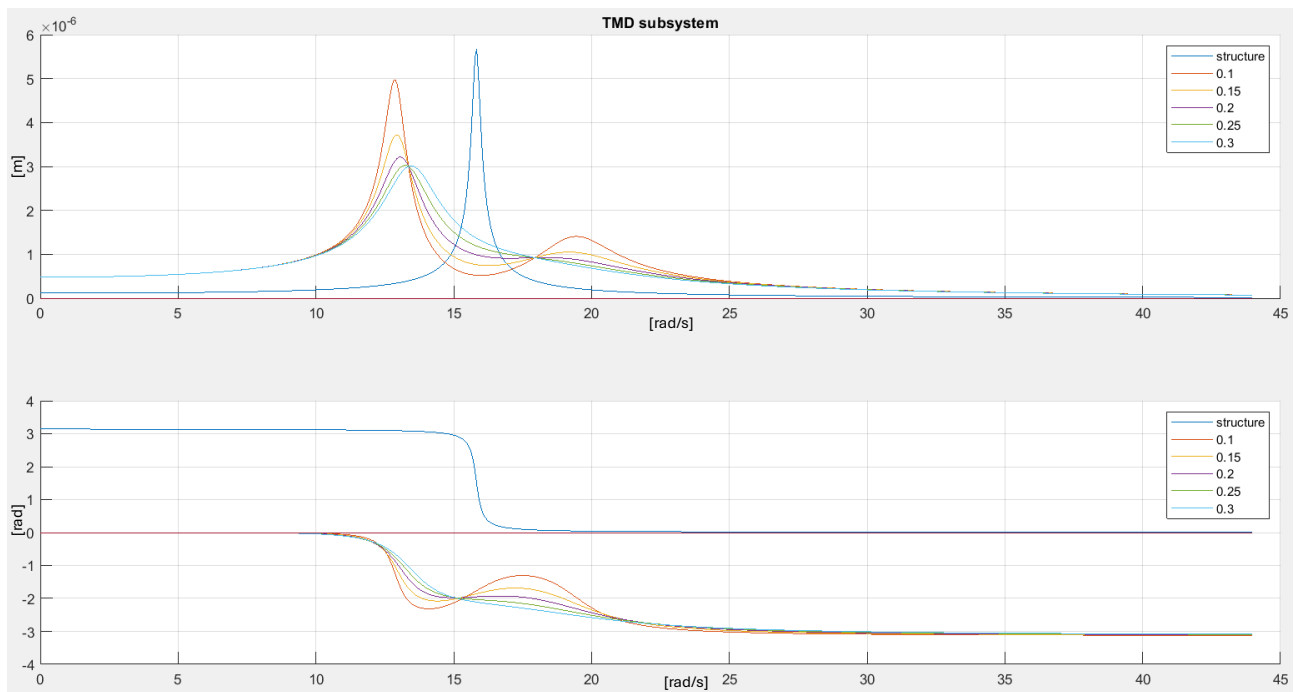
$$k_1 = 2.65 * 10^3 \frac{\text{N}}{\text{m}}$$

$$\omega_2 = \sqrt{\frac{k_2}{m_2}} = \omega_1 \quad c_2 = 2 * \xi_2 * m_2 * \omega_2$$

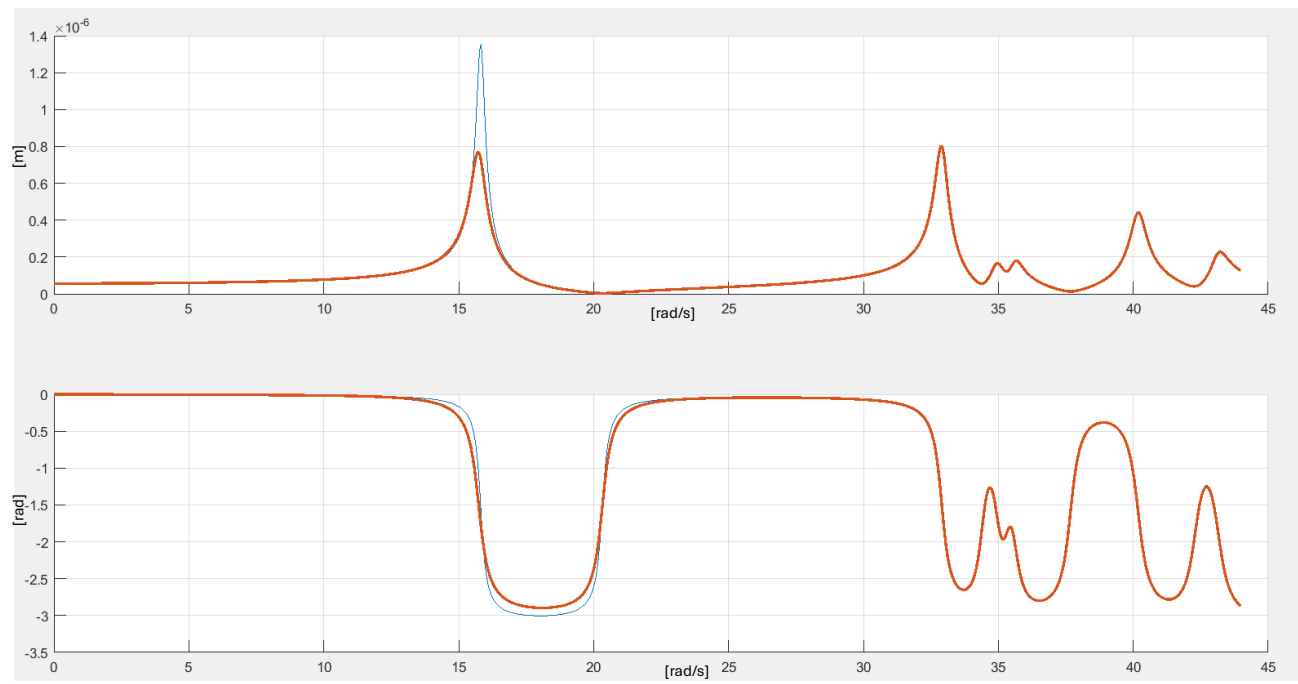
Setting the damping ratio to the limit value and varying the mass, we notice that if we increase m_2 up to the limit value the peak is slightly reduced but it moves to the left. So we chose an intermediate value $m_2 = 1500 \text{ kg}$ that maintains the peak near to the structure's one.



Then we modify the damping ratio and we notice that for small value we have two peaks, on the left and right to the structure's one; otherwise increasing the damping ratio the response 'slides' mainly to the first peak: we chose the max value allowed. The risk for a high damping ratio is to have a situation in which the two masses behave as one.



At this point the TMD was added to the entire structure, knowing that the best scenario in term of efficiency is when TMD is located in the antinode of the i -th mode shape, respectively in point B.



We obtain a reduction of the amplitude of the displacement in node a of: 40.8508%

Static deflection of the structure

The static deflection of the structure under the weight is computed by treating it as a distributed load with a constant value of $p_{0i} = m_i * g$, constant along element entire length.

Starting from EOMs for free dof:

$$[M_{FF}]\ddot{x}_F + [C_{FF}]\dot{x}_F + [K_{FF}]x_F = F_F^G$$

Imposing $\ddot{x} = \dot{x} = 0$

$$x_F = [K_{FF}]^{-1} F_F^G$$

To compute F_F^G :

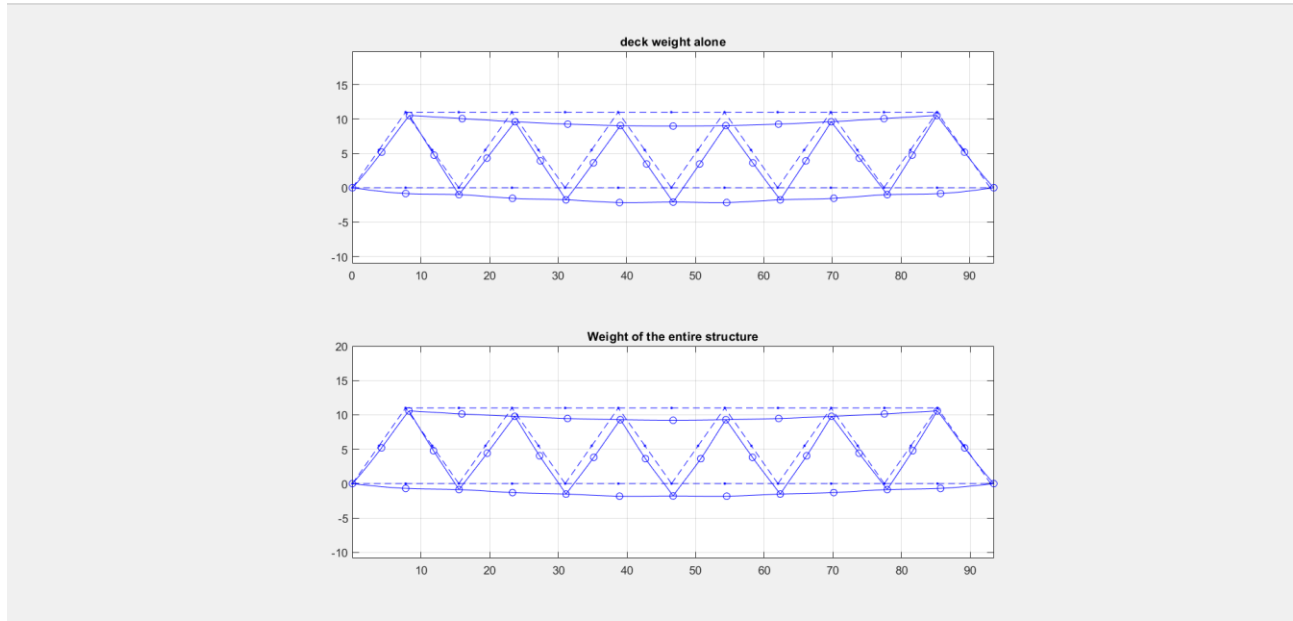
$$\delta W_{ext,k} = (\delta x_k^L)^T p_k \int_0^{L_k} f_w(\xi) d\xi = (\delta x_k)^T F_{k,distr}^L$$

In the case of computing the effect of the deck weight alone no coordinate transformation is needed because the local reference systems are aligned to the global one. Otherwise for the case of considering the weight of the entire structure we multiply the force vector for the rotation matrix $[\Lambda]$.

$$\delta W_{ext,k} = (\delta x_k^L)^T F_{k,distr}^L = (\delta x_k^G)^T [\Lambda] F_{k,distr}^L = (\delta x_k^G)^T F_k^G$$

At the end we multiply it for the extraction matrix:

$$\delta W = (\delta x)^T \sum_k [E_k]^T F_k^G = (\delta x)^T F^G \quad F^G = \begin{bmatrix} F_F^G \\ F_C + R \end{bmatrix}$$

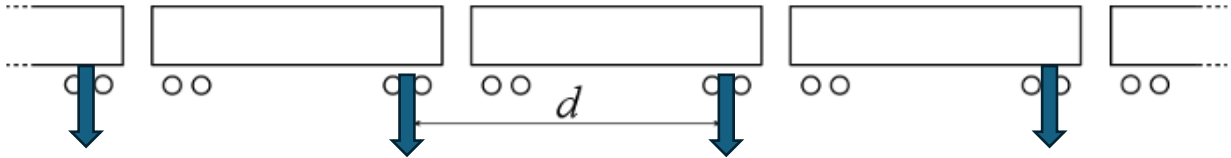


Plots are done with a scale factor of 40

The values of the maximum deflection are, respectively:

1. 0.036 m
2. 0.047 m

Critical velocities



In the case of a long sequence of moving loads we indicate with $\tau = \frac{d}{v}$ the periodicity of the excitation and with $\Omega_0 = \frac{2\pi}{\tau} = \frac{2\pi v}{d}$ the fundamental frequency of the generalized force. If the circular frequency $k\Omega_0$ of k -th harmonic component of the generalized force Q_i equals the beam's natural frequency ω_i , then the i -th mode enters in resonance:

$$k \frac{2\pi v}{d} = 2\pi f_i$$

$$V_{ik} = \frac{f_i d}{k}$$

Where:

$d = 27.5 \text{ m}$

k : integer value for $k \leq 3$ (generally the most dangerous situations correspond to low value of k)

$f_i \in (15,81 ; 34,9) \text{ Hz}$

we obtain for the first mode

$$V_{11} = 249,25 \text{ km/h}$$

$$V_{12} = 124.6 \text{ km/h}$$

$$V_{13} = 83.1 \text{ km/h}$$

for the second mode:

$$V_{21} = 518 \text{ km/h}$$

$$V_{22} = 259 \text{ km/h}$$

$$V_{23} = 172 \text{ km/h}$$

We also compute critical velocities also in the case of TMD. Obviously, the critical values for the second mode don't change because the TMD was designed to work around the first mode natural frequency.

With TMD, the first mode critical velocities become:

$$V_{13} = 237.2 \text{ km/h}$$

$$V_{12} = 118.6 \text{ km/h}$$

$$V_{11}^* = 79.1 \text{ km/h}$$